

UniDrag: Unified Multi-Field Prediction and Robust Shape Optimization for Vehicle Aerodynamics

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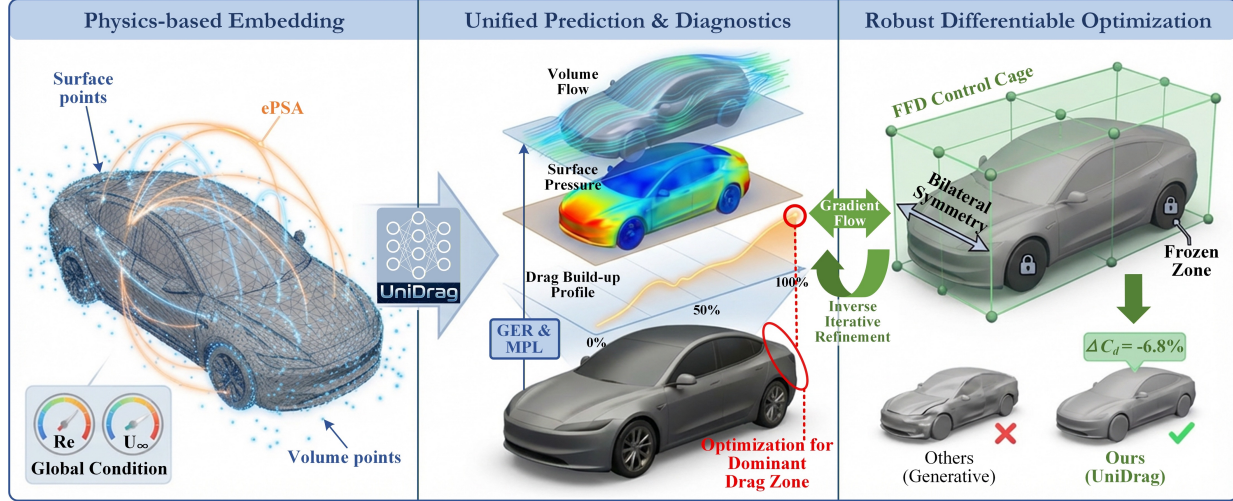


Figure 1. **UniDrag framework overview.** Given vehicle geometry, UniDrag predicts multi-field aerodynamic outputs and a streamwise build-up profile localizing dominant drag regions. At deployment, the frozen surrogate serves both as a predictor and an optimizer, enabling constrained shape refinement via FFD that achieves 6.8% drag reduction while satisfying engineering constraints.

Abstract

High-fidelity vehicle aerodynamics analysis is bottlenecked by costly CFD simulations. Neural surrogates accelerate prediction but lack inverse design capabilities, while existing generative optimization methods suffer from unstable convergence and frequent engineering constraint violations. We present UniDrag, a unified framework bridging multi-field aerodynamic prediction with robust differentiable shape optimization. Given a vehicle geometry, UniDrag predicts surface pressure, volume flow fields, drag coefficient C_d , and a streamwise build-up profile localizing drag contributions. Our architecture combines en-

hanced Physics-Sliced Attention (ePSA), Gated Expert Routing, and Modality-Protected Learning to prevent negative transfer across output modalities. At deployment, the frozen surrogate enables gradient-based optimization via Free-Form Deformation with engineering constraints. We introduce Expectation-over-Transformation to prevent adversarial exploitation of surrogate fragility. We curate a large-scale dataset of 15,000 vehicle geometries spanning four body types with GPU-accelerated LBM simulations. On this benchmark, UniDrag achieves C_d prediction R^2 of 0.937 (+7.6% over baselines) and 13.7% mean CFD-verified drag reduction with 100% success rate and only 21.3 mm average displacement (0.46% vehicle length).

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1. Introduction

High-fidelity computational fluid dynamics (CFD) remains the gold standard for vehicle aerodynamic analysis, yet each simulation consumes hours of computation. Modern automotive workflows demand rapid evaluation across thousands of geometric variants, rendering brute-force CFD

exploration prohibitively expensive (Tu et al., 2023).

This bottleneck has driven significant interest in learning-based surrogates. Recent neural operators, including Transolver (Wu et al., 2024), GeoTransolver (Adams et al., 2025), AB-UPT (Alkin et al., 2025), and DoMINO (Ranade et al., 2025), achieve competitive accuracy on benchmarks like DrivAerML (Ashton et al., 2024). However, these approaches are fundamentally prediction-only: they estimate flow fields and drag coefficients but provide no mechanism for inverse design (Wu et al., 2024; Adams et al., 2025; Alkin et al., 2025; Ranade et al., 2025). When a designer asks “how should I modify this shape to reduce drag?”, these methods remain silent.

A parallel line explores physics-guided shape optimization. Generative methods like PhysGen (You et al., 2025) and TripOptimizer (Vatani et al., 2025) integrate aerodynamic objectives into generative pipelines. Yet these approaches suffer from a critical limitation: lack of controllability. Without explicit geometric constraints, they frequently produce asymmetric bodies, unrealistic deformations, or shapes that exploit surrogate blind spots rather than achieving genuine aerodynamic improvement, often failing basic engineering requirements. We argue that practical vehicle design demands controllable refinement within strict engineering constraints (bilateral symmetry, region preservation, bounded deformations) rather than unconstrained generation.

We present **UniDrag**, a unified framework bridging multi-field aerodynamic prediction with robust differentiable shape optimization. As illustrated in Figure 1, UniDrag comprises two tightly coupled components: **(1) Train-once multi-field surrogate**. A single model jointly predicts surface pressure, volumetric flow fields, drag coefficient C_d , and a novel streamwise build-up profile that localizes where drag originates, providing actionable diagnostics for design decisions. The architecture combines enhanced Physics-Sliced Attention (ePSA), Gated Expert Routing (GER), and Modality-Protected Learning (MPL) to enable stable multi-task learning across heterogeneous outputs. **(2) Deploy-and-optimize pipeline**. The frozen surrogate enables gradient-based shape refinement via Free-Form Deformation (FFD) under explicit engineering constraints (bilateral symmetry, region freezing, bounded deformations). We introduce Expectation-over-Transformation (EoT) robust optimization to prevent adversarial deformations that exploit surrogate fragility.

Our contributions are:

- Unified prediction-optimization framework: the first approach bridging multi-field aerodynamic prediction with constraint-aware differentiable inverse design in a single pipeline.
- Scalable architecture: enhanced Physics-Sliced Atten-

tion (ePSA) with Gated Expert Routing and Modality-Protected Learning enables stable multi-task learning on large-scale CFD data.

- Robust optimization: Expectation over Transformation sampling with FFD constraints prevents adversarial exploitation of surrogate fragility.
- Large-scale benchmark: 15,000 vehicle geometries across four body types with LBM simulations.

On this benchmark, UniDrag achieves C_d prediction R^2 of 0.937 (+7.6% over baselines) and 13.7% mean CFD-verified drag reduction with 100% success rate and only 21.3 mm average displacement (0.46% vehicle length), satisfying strict engineering constraints where generative methods fail.

2. Related Work

Neural Surrogates for PDE and Aerodynamics. Neural operators (Li et al., 2020a;b; 2023) have emerged as fast alternatives to numerical PDE solvers, though early methods assume uniform grids or struggle with long-range interactions. For automotive aerodynamics, Transformer-based architectures achieve stronger performance: Transolver (Wu et al., 2024) introduces physics-sliced attention with linear complexity, GeoTransolver (Adams et al., 2025) adds multi-scale context, and AB-UPT (Alkin et al., 2025) couples surface-volume prediction. Despite strong accuracy, these are prediction-only architectures lacking inverse design. UniDrag extends this line with unified multi-field prediction and robustness mechanisms for optimization.

Geometric Deep Learning. A variety of techniques have been developed for learning on irregular geometries under the umbrella of geometric deep learning (Monti et al., 2017). Graph neural networks (Corso et al., 2024; Dwivedi et al., 2023) employ message-passing on connected graphs, while PointNet (Qi et al., 2017) and Point Transformer (Zhao et al., 2021) process unstructured point clouds directly. However, these methods are primarily designed for computer vision and graphics tasks, where the objective differs fundamentally from PDE solving. UniDrag builds upon Transolver’s physics-aware slicing mechanism (Wu et al., 2024), which learns to capture underlying physical structures rather than purely geometric features, and further enhances it with local geometry injection and 3D rotary position encoding (Su et al., 2024) for improved spatial reasoning on complex vehicle meshes.

Learning-Based Shape Optimization. Traditional aerodynamic optimization couples adjoint-based sensitivity analysis with CFD solvers (Africa et al., 2024), but inherits its prohibitive computational costs (Samareh, 2004). Recent learning-based approaches bypass explicit simulation:

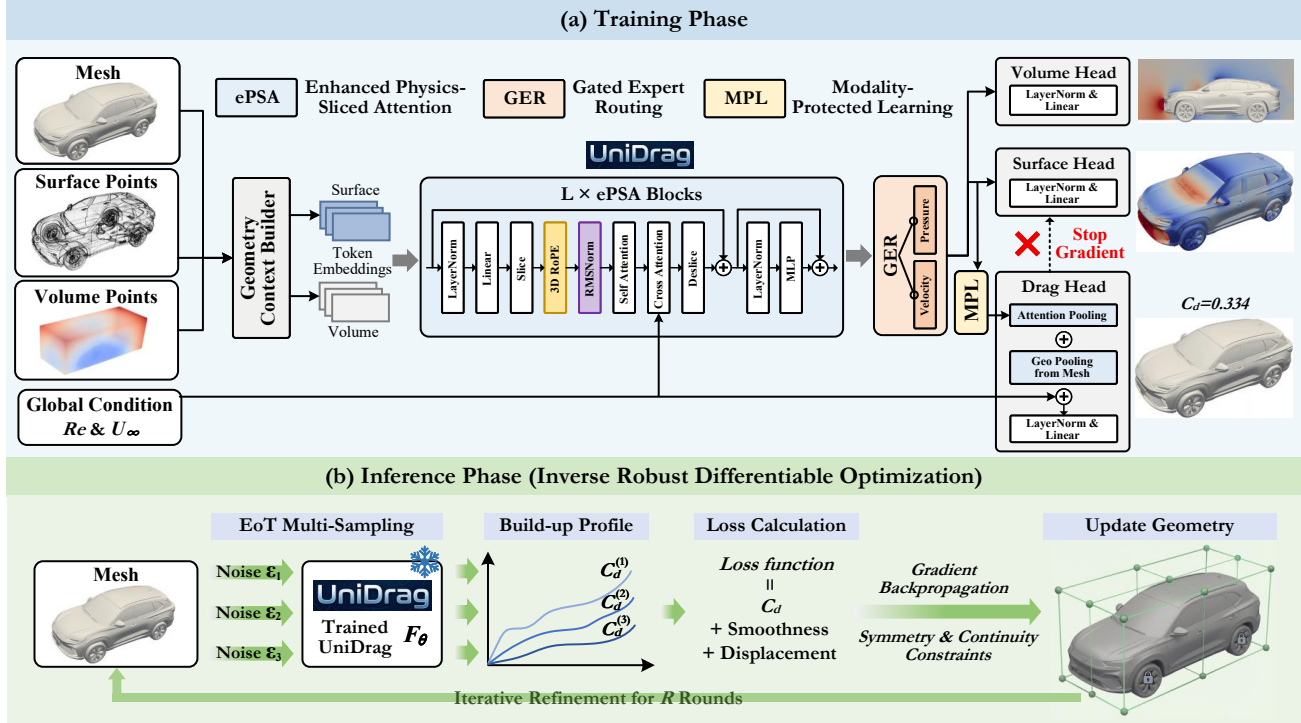


Figure 2. Overview of UniDrag. (a) Training: geometry-context tokens from surface and volume points are processed by L enhanced Physics-Sliced Attention (ePSA) blocks with gated expert routing (GER) and modality-protected learning (MPL), producing volume/surface field heads and a drag head. (b) Inference: inverse robust differentiable optimization performs multi-sampling, computes drag build-up profiles and an objective with smoothness/symmetry/continuity constraints, and backpropagates gradients to iteratively update the geometry.

TripOptimizer (Vatani et al., 2025) performs gradient updates in VAE latent space, PhysGen (You et al., 2025) combines flow-matching generation with physics-guided refinement, and AeroAgent (Elrefaie et al., 2025) integrates vision-language models with physics feedback. However, these methods lack engineering constraints (producing asymmetric or implausible shapes), suffer from latent-space drift, and show fake C_d reductions that vanish under CFD re-verification. UniDrag addresses these gaps through FFD-based differentiable optimization (Samareh, 2004) with explicit symmetry constraints, smoothness regularization, and Expectation-over-Transformation sampling that ensures CFD-verified drag reduction.

Automotive Aerodynamics Datasets. Existing public datasets for vehicle aerodynamics are limited in scale and diversity. DrivAerNet (Elrefaie et al., 2024a) provides 4,000 samples from parametric morphing of a single base geometry; DrivAerNet++ (Elrefaie et al., 2024b) extends to 8,000 samples across three base topologies. These datasets share critical limitations: restricted topology variation limits generalization, and none support CFD re-verification of optimized geometries essential for validating inverse design. To support both prediction and optimization evaluation, we curate a dataset of 15,000 vehicles spanning four

body types (Sedan, SUV, Sport Car, MPV) with complete CFD labels and reproducible simulation configurations for re-verification.

3. Method

3.1. Problem Setup

We study a unified learning and optimization problem for vehicle aerodynamics, where a differentiable model maps geometry to multi-scale aerodynamic quantities and supports inverse design. As shown in Figure 2, UniDrag couples a *training phase* (a)—where geometry-context tokens are processed by L ePSA blocks and routed by GER/MPL to Volume, Surface, and Drag heads—with an *inference phase* (b), in which the frozen surrogate $\mathcal{F}_\theta$ is queried under EoT multi-sampling and the resulting drag build-up profiles, together with smoothness, symmetry, and continuity constraints, are backpropagated through FFD to iteratively refine the geometry.

Let $G \in \mathcal{G}$ denote a vehicle exterior geometry, where $\mathcal{G}$ represents the space of admissible vehicle shapes. During deployment, the geometry is parameterized by a differentiable deformation operator $T_\phi(\cdot)$, yielding

$$G_\phi = T_\phi(G), \quad (1)$$

where ϕ represents the deformation parameters. We define two query point sets. Surface points are denoted by $X_s = \{\mathbf{x}_i^s\}_{i=1}^{N_s}$, sampled on the vehicle surface, each associated with an outward normal $\mathbf{n}_i^s \in \mathbb{R}^3$. Volume points are denoted by $X_v = \{\mathbf{x}_j^v\}_{j=1}^{N_v}$, distributed in the surrounding flow domain. A unified surrogate model

$$\mathcal{F}_\theta : (G, X_s, X_v, c) \mapsto (\hat{f}_s, \hat{f}_v, \widehat{C_d}, \hat{b}) \quad (2)$$

with learnable parameters θ is trained to jointly predict surface fields $\hat{f}_s = \{\hat{f}_s(\mathbf{x}_i^s)\}_{i=1}^{N_s}$, volumetric flow fields $\hat{f}_v = \{\hat{f}_v(\mathbf{x}_j^v)\}_{j=1}^{N_v}$, the global drag coefficient $\widehat{C_d}$, and a streamwise drag build-up profile $\hat{b} \in \mathbb{R}^K$, where $K = 100$ denotes the number of streamwise segments used to discretize the drag distribution. Here, c represents global flow conditions such as freestream velocity and Reynolds number.

3.2. Global Interaction via Enhanced PSA

Vehicle aerodynamics involves long-range spatial dependencies: front-body geometry affects rear wake structures, and roof curvature influences overall flow separation (Elrefaie et al., 2024a;b). Standard point-wise operators fail to capture such global interactions efficiently. Physics-Sliced Attention (PSA) (Wu et al., 2024) addresses this by softly routing N point tokens to S learnable slices ($S \ll N$), performing attention at the slice level, and scattering information back, reducing complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(NS + S^2)$.

However, vanilla PSA lacks three properties critical for CFD data: (1) awareness of local surface geometry for routing decisions, (2) training stability under large magnitude variations in pressure/velocity fields, and (3) 3D spatial encoding for slice-level attention. We propose enhanced PSA (ePSA) with the following improvements.

Local Geometry Injection. For each point i , we aggregate and standardize neighborhood features within a local radius r :

$$\mathbf{g}_i = \frac{\psi(\mathcal{N}_r(i)) - \mu_{\mathcal{N}}}{\sigma_{\mathcal{N}} + \epsilon}, \quad \mathbf{h}_i^{\text{aug}} = \text{Concat}[\mathbf{h}_i, \mathbf{g}_i], \quad (3)$$

where $\psi(\cdot)$ encodes local curvature and normals, and standardization ensures scale-invariant encoding across vehicle regions with varying geometric complexity.

Soft Slice Routing. Point tokens are assigned to S physics slices via learned routing:

$$\pi_{ik} = \text{softmax}_k(\mathbf{w}_k^\top \mathbf{h}_i^{\text{aug}}), \quad (4)$$

and slice tokens with their spatial centers are obtained by:

$$\mathbf{s}_k = \sum_{i=1}^N \pi_{ik} \mathbf{W}_v \mathbf{h}_i^{\text{aug}}, \quad \mathbf{p}_k = \frac{\sum_i \pi_{ik} \mathbf{x}_i}{\sum_i \pi_{ik}}. \quad (5)$$

QK-RMSNorm and 3D-RoPE. Before slice attention, we apply RMS normalization (Zhang & Sennrich, 2019) to query and key vectors to prevent score explosions from CFD’s large dynamic range. We then incorporate slice center positions $\mathbf{p}_k$ via 3D rotary embeddings, encoding that upstream slices causally affect downstream ones. The attention is:

$$\mathbf{s}' = \text{softmax}\left(\frac{\mathbf{q}^{\text{rot}}(\mathbf{k}^{\text{rot}})^\top}{\sqrt{d}}\right) \mathbf{v}, \quad (6)$$

where d is the attention head dimension. Finally, updated slice information is scattered back: $\mathbf{h}'_i = \sum_k \pi_{ik} \mathbf{s}'_k$.

3.3. Task Processing with Gated Expert Routing

The globally informed point features produced by ePSA are shared across heterogeneous prediction tasks, including surface field estimation, volumetric flow prediction, and global aerodynamic coefficients. These tasks differ significantly in their physical semantics, spatial support, and learning difficulty. Using a single shared transformation for all tasks often leads to suboptimal performance due to representation interference.

To address this issue, we introduce Gated Expert Routing (GER), which enables task-adaptive feature processing through a mixture-of-experts architecture. Given the updated point features $\mathbf{h}'_i \in \mathbb{R}^d$, each token is routed to a set of M_e expert networks via a learnable gating function:

$$\alpha_i = \text{softmax}(\mathbf{W}_g \mathbf{h}'_i), \quad (7)$$

where $\alpha_i \in \mathbb{R}^{M_e}$ denotes the routing weights over experts, and $\mathbf{W}_g$ is a learnable projection. Each expert $\mathcal{E}_m(\cdot)$ applies a task-specialized transformation to the input features. The routed output is obtained by a weighted combination of expert outputs:

$$\tilde{\mathbf{h}}_i = \sum_{m=1}^{M_e} \alpha_{i,m} \mathcal{E}_m(\mathbf{h}'_i). \quad (8)$$

The resulting features $\tilde{\mathbf{h}}_i$ preserve the global contextual information from ePSA while allowing different experts to focus on distinct physical aspects of the problem. This adaptive routing mechanism enables effective capacity allocation across heterogeneous tasks, forming the basis for subsequent modality-specific prediction heads.

3.4. Modality-Protected Learning

After ePSA and GER, each point has a globally informed, task-adaptive feature $\tilde{\mathbf{h}}_i$. However, joint training on heterogeneous outputs—surface fields, volume fields, and global scalars—risks negative transfer. Low-dimensional objectives (C_d , build-up) converge faster and can dominate gradients, causing the backbone to take shortcuts that sacrifice

field accuracy. We introduce Modality-Protected Learning (MPL) to control gradient flow across modalities based on physical causality.

Features are partitioned into surface features $\tilde{\mathbf{h}}^{(s)}$ and volume features $\tilde{\mathbf{h}}^{(v)}$. We design information flow to respect physical dependencies:

Surface $\rightarrow$ Surface (direct). Surface pressure is the primary physical quantity determined by geometry. We predict it directly without gradient isolation:

$$\hat{f}_s = \text{SurfaceHead}(\tilde{\mathbf{h}}^{(s)}). \quad (9)$$

Surface $\rightarrow$ Volume (forward only). Volume velocity in the near-wake region depends on upstream surface conditions (boundary layer separation, pressure gradients). Volume features receive surface context, but volume prediction errors should not corrupt surface representations:

$$\hat{f}_v = \text{VolumeHead}(\tilde{\mathbf{h}}^{(v)}, \text{stopgrad}(\tilde{\mathbf{h}}^{(s)})). \quad (10)$$

Surface $\rightarrow$ Global Scalars (partial isolation). Drag coefficient C_d and the 100-segment build-up profile $\hat{b}$ are physically determined by surface stress integration. The drag head receives three inputs: (1) attention-pooled surface features with gradient isolation, (2) geometry encoding from a dedicated encoder, and (3) global flow conditions (Reynolds number Re , freestream velocity U_∞):

$$(\widehat{C}_d, \hat{b}) = \text{DragHead}(\text{stopgrad}(\text{AttnPool}(\tilde{\mathbf{h}}^{(s)})), \mathbf{z}_{\text{geo}}, \mathbf{c}). \quad (11)$$

Here stopgrad is applied only to the surface features, while geometry encoding $\mathbf{z}_{\text{geo}}$ and global conditions $\mathbf{c}$ allow normal gradient flow. This design prevents the strong C_d gradient from hijacking the shared point-wise encoder, while still enabling the geometry encoder and condition embedding to learn from scalar supervision. The drag head thus benefits from rich surface representations without corrupting them, maintaining field prediction accuracy.

3.5. Training Objectives

UniDrag is trained end-to-end on dataset $\mathcal{D} = \{(G, f_s, f_v, C_d, b)\}$. Surface and volume fields are supervised by relative L_2 losses to normalize across different physical scales:

$$\mathcal{L}_s = \frac{\|\hat{f}_s - f_s\|_2}{\|f_s\|_2}, \quad \mathcal{L}_v = \frac{\|\hat{f}_v - f_v\|_2}{\|f_v\|_2}. \quad (12)$$

Global quantities use normalized losses on C_d and the build-up profile:

$$\mathcal{L}_{C_d} = \frac{|\widehat{C}_d - C_d|}{C_d}, \quad \mathcal{L}_b = \frac{\|\hat{b} - b\|_2}{\|b\|_2}. \quad (13)$$

By normalizing each loss to relative error, all objectives operate at comparable scales (typically 0.1–0.3 during training). Combined with MPL’s gradient isolation (§3.4), this eliminates the need for manual loss balancing. The overall objective is:

$$\mathcal{L} = \mathcal{L}_s + \mathcal{L}_v + \mathcal{L}_{C_d} + \mathcal{L}_b, \quad (14)$$

with all weights set to 1. Gradients respect the modality-protected constraints: surface field loss updates the full backbone, while C_d and build-up losses only affect the geometry encoder and condition embedding.

3.6. Robust Differentiable Optimization

At deployment, we freeze the trained surrogate $\mathcal{F}_\theta$ and optimize vehicle geometry to minimize drag. The key challenge is that neural surrogates can be exploited by adversarial deformations—geometries that appear to reduce predicted C_d but fail CFD verification. We introduce Robust Differentiable Optimization (RDO) to ensure physically valid, manufacturable shape modifications.

FFD Parameterization. We use Free-Form Deformation (FFD) (Samareh, 2004) to constrain geometry changes to a smooth, low-dimensional space. A $5 \times 5 \times 5$ control lattice $\{\mathbf{P}_{ijk}\}$ is placed around the vehicle, and surface points are deformed via trivariate Bernstein polynomials (Braess & Sauer, 2004):

$$T_\phi(\mathbf{x}) = \sum_{i,j,k} B_i^{(l)}(s) B_j^{(m)}(t) B_k^{(n)}(u) \mathbf{P}_{ijk}, \quad (15)$$

where $(s, t, u) \in [0, 1]^3$ are local coordinates and $\phi = \{\Delta \mathbf{P}_{ijk}\}$ are learnable control point displacements. FFD guarantees C^2 continuous deformations by construction.

Bilateral Symmetry. Vehicles must remain left-right symmetric. We only optimize control points on one side ($y > 0$), and mirror displacements to the other side: $\Delta \mathbf{P}_{i,j,k} = (\Delta x, -\Delta y, \Delta z)$ for $y < 0$.

Region-Specific Optimization. Rather than deforming the entire vehicle, we target aerodynamically significant regions identified by the build-up profile. For each region (front 15%, rear 15%, roof 20%), we apply a C^3 -continuous weight function that smoothly transitions from full deformation to zero at boundaries:

$$w(t) = [35t^4 - 84t^5 + 70t^6 - 20t^7]^2, \quad (16)$$

ensuring deformations vanish at region boundaries without discontinuities. Wheels, windows, and underbody are frozen.

Expectation-over-Transformation (EoT). To prevent adversarial exploitation, we optimize under stochastic perturbations. At each step, we sample M random point subsets and

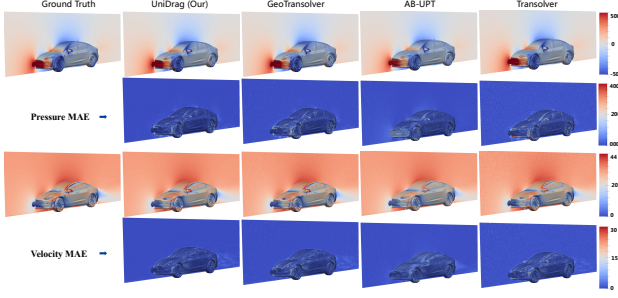


Figure 3. Field prediction comparison. UniDrag achieves the lowest error in both surface and volume fields, accurately capturing high-gradient regions (A-pillar, rear wake) where baselines show larger deviations.

average predictions:

$$\bar{C}_d = \frac{1}{M} \sum_{m=1}^M \mathcal{F}_\theta(T_\phi(G); \epsilon_m), \quad (17)$$

where ϵ_m represents random sampling of 50k surface points. This smooths the loss landscape and suppresses deformations that exploit specific point configurations.

Smoothness and Displacement Constraints. We regularize control point displacements to ensure manufacturable shapes:

$$\mathcal{R} = \lambda_{C2} \|\nabla^2 \phi\|^2 + \lambda_{C3} \|\nabla^3 \phi\|^2 + \lambda_{nb} \sum_{i \sim j} \|\phi_i - \phi_j\|^2, \quad (18)$$

where ∇^2 and ∇^3 are discrete second and third-order differences penalizing non-smooth control point configurations. A displacement penalty $\mathcal{P} = \max(0, \|\Delta \mathbf{x}\|/d_{\max} - 1)^2$ enforces maximum deformation bounds (typically 3% of vehicle length).

Optimization Objective. The final objective combines drag minimization with constraints:

$$\min_{\phi} \bar{C}_d + \mathcal{R} + \lambda_{\text{disp}} \mathcal{P}. \quad (19)$$

We optimize using Adam (Zhang, 2018; Loshchilov & Hutter, 2017) with learning rate 0.01 for 30–60 steps, with gradient clipping for stability. The entire pipeline is end-to-end differentiable: gradients flow from C_d through the frozen surrogate back to FFD control points, enabling efficient shape optimization without CFD-in-the-loop.

4. Experiments

We design comprehensive experiments to validate UniDrag on both multi-field prediction accuracy and inverse design effectiveness.

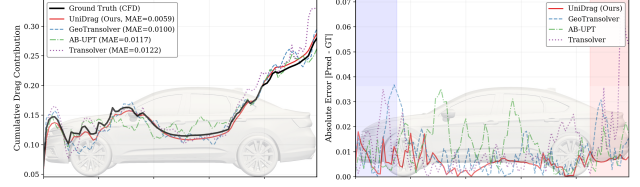


Figure 4. Drag build-up profile prediction for a sedan. Left: Cumulative drag contribution across 100 streamwise segments with vehicle silhouette as spatial reference. UniDrag (MAE=0.0059) best matches CFD ground truth, with largest drag increases at front and rear regions. Right: Per-segment absolute error. UniDrag maintains consistently low errors while baselines show higher fluctuations in critical regions.

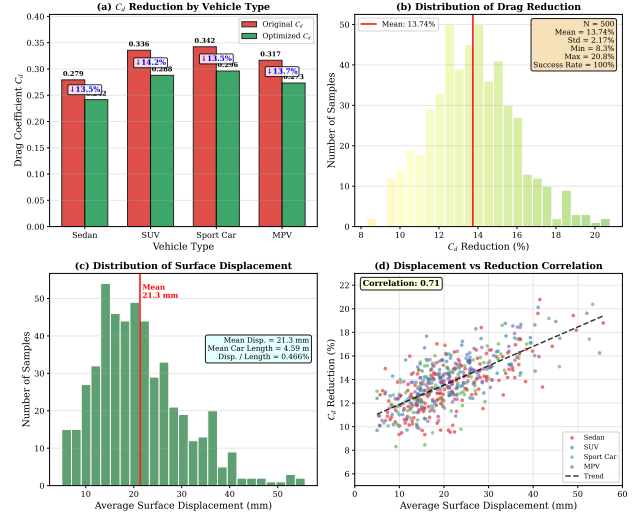


Figure 5. CFD verification of optimization effectiveness on 500 vehicles. (a) Consistent C_d reduction (13–14%) across all vehicle types. (b) 100% success rate with mean reduction of 13.73%. (c) Minimal surface displacement (mean 21.3 mm, 0.46% of vehicle length). (d) Positive correlation between displacement and reduction, indicating that larger shape modifications yield greater aerodynamic improvements.

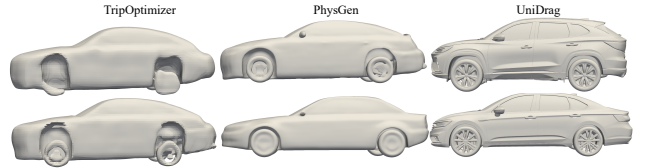


Figure 6. Qualitative comparison of optimized geometries. TripOptimizer: Severe mesh distortion, unsuitable for CFD. PhysGen: Over-smoothed with lost details. UniDrag: Preserves topology with C^2/C^3 -continuous, manufacturable deformations.

4.1. Experimental Setup

Dataset. We curate a large-scale dataset comprising 15,000 distinct vehicle geometries across four body classes (Sedan, SUV, Sport Car, MPV), simulated via GPU-accelerated Lattice-Boltzmann Method (LBM) (Chen & Doolen, 1998). The data are split into 12,000/1,500/1,500

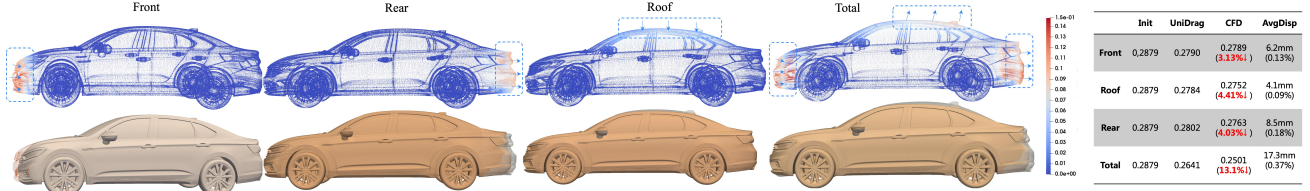


Figure 7. Region-specific optimization visualization. Top: Wireframe with deformation magnitude. Bottom: Rendered geometries. Right: CFD-verified results showing C_d reduction and minimal displacement for each region. Total configuration achieves 13.1% reduction with only 17.3mm avg. displacement.

for training/validation/testing, with 500 test samples reserved for CFD-verified optimization evaluation. Each sample includes surface pressure, volume velocity fields, and the 100-segment drag build-up profile. Detailed dataset construction, LBM simulation parameters, and comparison with existing benchmarks are provided in Appendix D.

Baselines. For prediction, we compare against Transolver (Wu et al., 2024), GeoTransolver (Adams et al., 2025), and AB-UPT (Alkin et al., 2025). For inverse design, we compare against PhysGen (You et al., 2025) and TripOptimizer (Vatani et al., 2025). Detailed baseline implementations and hyperparameter configurations are provided in Appendix F.

Evaluation Metrics. We evaluate UniDrag across two dimensions: (1) Prediction: R^2 and Relative L_1/L_2 for C_d , surface pressure, and volume velocity fields. Build-up R^2 evaluates the 100-segment drag profile prediction. (2) Optimization: Success rate (percentage of samples with C_d reduction), mean/max C_d reduction ratio, average surface displacement, and displacement as percentage of vehicle length. Detailed metric definitions are provided in Appendix F.

Implementation Details. UniDrag consists of 12 ePSA blocks with a 384-dimensional hidden space; the last 4 blocks include GER with 3 experts each, while the first 8 blocks use standard FFN. We train for 100 epochs using AdamW optimizer (Loshchilov & Hutter, 2017) with cosine learning rate scheduling (Li & Arora, 2019) on 8× NVIDIA A100 GPUs. For FFD-based shape optimization (Samareh, 2004), we use a $5 \times 5 \times 5$ control grid with step size 0.01 and 30 optimization steps. The smoothness regularization weights are $\lambda_{C2} = 15$, $\lambda_{C3} = 8$, $\lambda_{nb} = 5$ for neighbor consistency, and $\lambda_{disp} = 500$ for displacement penalty. We use $M = 5$ EoT samples per optimization step. Detailed configurations are provided in Appendix C.

4.2. Prediction Performance

Quantitative Comparison and Field Visualization. As summarized in Table 1, UniDrag achieves SOTA performance across all predictive metrics. For drag coefficient prediction, UniDrag attains an R^2 of 0.937, representing a 7.58% relative improvement over the second-best Geo-

Table 1. UniDrag experimental results on surface pressure field, volume velocity field, and C_d . Lower relative L_2 and L_1 are better for fields, higher R^2 is better for C_d . Best results are bold, second-best underlined. “Promotion” refers to our method’s relative error reduction or gain w.r.t. the second-best model.

Model	Surface Pressure		Volume Velocity		C_d
	Rel L_2 ↓	Rel L_1 ↓	Rel L_2 ↓	Rel L_1 ↓	R^2 ↑
Transolver (Wu et al., 2024)	0.3340	0.2505	0.2780	0.2085	0.8230
AB-UPT (Alkin et al., 2025)	0.3050	0.2288	0.2310	0.1733	0.8590
GeoTransolver (Adams et al., 2025)	0.2970	0.2228	0.2110	0.1583	0.8710
UniDrag (Ours)	0.2620	0.1965	0.1850	0.1388	0.9370
Relative Promotion	11.8%	11.8%	12.3%	12.3%	7.58%

Transolver. For field-level predictions, UniDrag reduces surface pressure Rel- L_2 error by 11.8% (0.297→0.262) and volume velocity error by 12.3% (0.211→0.185), demonstrating effective capture of both integral quantities and local flow features. Fig. 3 visualizes the predicted fields, where UniDrag produces accurate pressure and velocity reconstructions with minimal error in aerodynamically critical high-gradient regions such as the A-pillar junction and rear wake separation zone.

Drag Build-up Profile Prediction. Fig. 4 visualizes the 100-segment drag build-up profile for a sedan. UniDrag (red) closely tracks CFD ground truth (black), achieving MAE of 0.0059 vs. GeoTransolver (0.0100), AB-UPT (0.0117), and Transolver (0.0122). The build-up curve shows drag accumulates most rapidly at front and rear regions. UniDrag maintains consistently low errors across all segments while baselines exhibit larger fluctuations.

4.3. Optimization Performance

Optimization Effectiveness and CFD Verification. To validate that UniDrag-optimized shapes achieve real aerodynamic improvements, we re-simulate 500 optimized geometries using high-fidelity CFD. The optimization targets three aerodynamically critical regions: front 15% (hood/fascia), rear 15% (trunk/diffuser), and top 20% (roof). As shown in Fig. 5, CFD verification confirms 100% success rate: all 500 samples achieve genuine C_d reduction, demonstrating that our surrogate-guided optimization reliably transfers to ground-truth physics. The mean CFD-verified reduction reaches 13.73%, consistent across vehicle types:

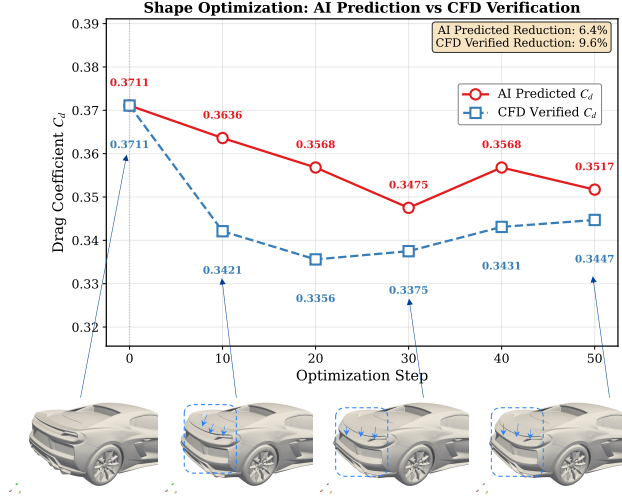


Figure 8. Optimization trajectory over 50 steps. AI-predicted and CFD-verified C_d curves show consistent trends, confirming reliable gradient guidance. CFD verification achieves 9.6% reduction, validating that surrogate-guided optimization transfers to real physics.

Sedan (13.5%), SUV (14.2%), Sport Car (13.5%), and MPV (13.7%). These improvements require only 21.3 mm average displacement (0.46% of vehicle length), confirming that our robust differentiable optimization produces manufacturable shape changes.

Comparison with Generative Baselines. Fig. 6 compares the optimized geometries from generative baselines (TripOptimizer, PhysGen) and our FFD-based method. TripOptimizer produces severely distorted meshes with missing wheel arches, non-physical surface discontinuities, and broken topology, rendering the output unsuitable for CFD simulation or manufacturing. PhysGen generates smoother but overly simplified shapes that lose critical geometric details (e.g., mirrors, grilles, wheel spokes), making them impractical for engineering analysis. In contrast, UniDrag preserves the original vehicle topology while applying smooth, localized deformations with explicit C^2/C^3 continuity constraints, ensuring curvature-continuous surfaces suitable for aerodynamic analysis. Our FFD-based approach guarantees manufacturable outputs directly compatible with CFD validation, enabling a closed-loop design workflow.

Region-Specific Optimization Visualization. Fig. 7 visualizes the optimization results for different aerodynamic regions on a representative sedan. We target four configurations: Front (15%), Rear (15%), Roof (20%), and Total (all regions combined). The top row displays wireframe overlays highlighting the deformation magnitude, while the bottom row shows the rendered optimized geometries. Importantly, all deformations satisfy strict engineering constraints: average displacements range from 4.1mm (Roof) to

17.3mm (Total), corresponding to less than 0.4% of vehicle length. The CFD-verified C_d reductions demonstrate that each region contributes meaningfully: Front (3.1%), Roof (4.4%), Rear (4.0%), with the Total configuration achieving the best result (13.1% reduction, from 0.288 to 0.250). The close agreement between UniDrag predictions and CFD verification (within 5% relative error) confirms that our robust optimization produces physically valid, manufacturable shape modifications.

Optimization Trajectory and CFD Consistency. Fig. 8 tracks the optimization trajectory over 50 iterations, comparing AI-predicted C_d (red) with CFD-verified C_d (blue) at each step. Both curves exhibit consistent downward trends, with the CFD-verified reduction (9.6%) closely matching the AI prediction (6.4%). Crucially, the two curves maintain similar variation patterns throughout the optimization process, demonstrating that our surrogate model provides reliable gradient guidance even under continuous shape deformation. This consistency validates that UniDrag does not exploit model blind spots but achieves genuine aerodynamic improvements that transfer to high-fidelity CFD. The bottom thumbnails illustrate the progressive shape modifications at key iterations.

4.4. Ablation Studies

We conduct ablation studies to verify the contribution of each UniDrag component.

Table 2. Ablation study of UniDrag prediction components.

Configuration	$C_d R^2 \uparrow$	Surface $L_2 \downarrow$	Volume $L_2 \downarrow$	Build-up $R^2 \uparrow$
Full UniDrag	0.937	0.262	0.185	0.971
w/o MPL (stop-grad)	0.898	0.295	0.212	0.945
w/o Local geometry injection	0.905	0.288	0.204	0.954
w/o 3D-RoPE	0.915	0.278	0.196	0.961
w/o GER (MoE-FFN)	0.921	0.271	0.191	0.965
w/o QK-RMSNorm	0.924	0.269	0.189	0.967
GeoTransolver baseline	0.871	0.297	0.211	0.951

As shown in Table 2, MPL contributes most significantly: removing gradient isolation causes 4.2% drop in $C_d R^2$ (0.937→0.898), indicating severe negative transfer between heterogeneous outputs. Local geometry injection provides 3.4% improvement by encoding scale-invariant curvature information for slice routing. 3D-RoPE enables spatial reasoning with 2.3% gain. GER allows task-specific specialization with 1.7% improvement. QK-RMSNorm stabilizes attention on irregular point clouds with 1.4% contribution.

Table 3. EoT sampling (M).

M	$\Delta C_d^{CFD} \downarrow$	Succ.
1	-11.2%	97%
3	-12.8%	99%
5	-13.7%	100%
7	-13.5%	100%

Table 4. FFD grid resolution.

Grid	$\Delta C_d^{CFD} \downarrow$
3^3	-6.8%
5^3	-13.7%
7^3	-15.2%
9^3	-16.1%

Tables 3–4 show optimization ablations on 50 samples. EoT with $M = 5$ achieves 100% success; finer FFD grids yield larger reduction but require stronger smoothness constraints.

5. Conclusion

We presented UniDrag, a unified framework bridging multi-field aerodynamic prediction with robust differentiable shape optimization. We curated a large-scale dataset of 15,000 vehicle geometries across four body types with GPU-accelerated LBM simulations. By combining enhanced Physics-Sliced Attention (ePSA), Gated Expert Routing, and Modality-Protected Learning, UniDrag achieves state-of-the-art prediction (C_d $R^2=0.937$) while enabling CFD-verified drag reduction (13.73% mean, 100% success rate) with minimal geometric modification (21.3mm avg. displacement). The streamwise build-up profile provides interpretable diagnostics, and FFD-based optimization with C^2/C^3 constraints ensures manufacturable outputs.

Limitations. Our dataset focuses on passenger vehicles; broader generalization requires domain-specific data. FFD parameterization ensures manufacturability but may limit design freedom. Current optimization targets drag only; multi-objective extension (lift, acoustics) remains future work.

Impact Statement

This work presents an end-to-end pipeline for aerodynamic vehicle design, combining high-accuracy surrogate prediction with robust closed-loop geometry optimization under CFD verification. By enabling faster, traceable design iteration, UniDrag can reduce engineering turnaround time and development costs, potentially contributing to improved energy efficiency in transportation. The approach may accelerate the adoption of low-drag vehicle designs, supporting sustainability goals in the automotive industry. We identify no significant ethical or societal risks beyond standard machine learning considerations.

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References

- Adams, C., Ranade, R., Cherukuri, R., and Choudhry, S. Geotransolver: Learning physics on irregular domains using multi-scale geometry aware physics attention transformer. *arXiv preprint arXiv:2512.20399*, 2025.
- Africa, P. C., Fumagalli, I., Bucelli, M., Zingaro, A., Fedele, M., Quarteroni, A., et al. lifex-cfd: An open-source computational fluid dynamics solver for cardiovascular applications. *Computer Physics Communications*, 296: 109039, 2024.
- Alkin, B., Bleeker, M., Kurle, R., Kronlachner, T., Sonnleitner, R., Dorfer, M., and Brandstetter, J. AB-UPT: Scaling neural CFD surrogates for high-fidelity automotive aerodynamics simulations via anchored-branched universal physics transformers. *arXiv preprint arXiv:2502.09692*, 2025.
- Ashton, N., Mockett, C., Fuchs, M., Fliessbach, L., Hetmann, H., Knacke, T., Schonwald, N., Skaperdas, V., Fotiadis, G., Walle, A., et al. Drivaerml: High-fidelity computational fluid dynamics dataset for road-car external aerodynamics. *arXiv preprint arXiv:2408.11969*, 2024.
- Braess, D. and Sauer, T. Bernstein polynomials and learning theory. *Journal of Approximation Theory*, 128(2):187–206, 2004.
- Cao, S., Chen, H., Chen, P., Cheng, Y., Cui, Y., Deng, X., Dong, Y., Gong, K., Gu, T., Gu, X., et al. Hunyuanimage 3.0 technical report. *arXiv preprint arXiv:2509.23951*, 2025.
- Chen, S. and Doolen, G. D. Lattice boltzmann method for fluid flows. *Annual review of fluid mechanics*, 30(1): 329–364, 1998.
- Corso, G., Stark, H., Jegelka, S., Jaakkola, T., and Barzilay, R. Graph neural networks. *Nature Reviews Methods Primers*, 4(1):17, 2024.
- Dwivedi, V. P., Joshi, C. K., Luu, A. T., Laurent, T., Bengio, Y., and Bresson, X. Benchmarking graph neural networks. *Journal of Machine Learning Research*, 24 (43):1–48, 2023.
- Elrefaie, M., Dai, A., and Ahmed, F. Drivaernet: A parametric car dataset for data-driven aerodynamic design and graph-based drag prediction. In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, volume 88360, pp. V03AT03A019. American Society of Mechanical Engineers, 2024a.

- Elrefaie, M., Morar, F., Dai, A., and Ahmed, F. Drivaer-net++: A large-scale multimodal car dataset with computational fluid dynamics simulations and deep learning benchmarks. *Advances in Neural Information Processing Systems*, 37:499–536, 2024b.
- Elrefaie, M., Qian, J., Wu, R., Chen, Q., Dai, A., and Ahmed, F. AI agents in engineering design: a multi-agent framework for aesthetic and aerodynamic car design. In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, volume 89237, pp. V03BT03A048. American Society of Mechanical Engineers, 2025.
- Lai, Z., Zhao, Y., Liu, H., Zhao, Z., Lin, Q., Shi, H., Yang, X., Yang, M., Yang, S., Feng, Y., et al. Hunyuan3d 2.5: Towards high-fidelity 3d assets generation with ultimate details. *arXiv preprint arXiv:2506.16504*, 2025.
- Li, Z. and Arora, S. An exponential learning rate schedule for deep learning. *arXiv preprint arXiv:1910.07454*, 2019.
- Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A., and Anandkumar, A. Fourier neural operator for parametric partial differential equations. *arXiv preprint arXiv:2010.08895*, 2020a.
- Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A., and Anandkumar, A. Neural operator: Graph kernel network for partial differential equations. *arXiv preprint arXiv:2003.03485*, 2020b.
- Li, Z., Huang, D. Z., Liu, B., and Anandkumar, A. Fourier neural operator with learned deformations for PDEs on general geometries. *Journal of Machine Learning Research*, 24(388):1–26, 2023.
- Loshchilov, I. and Hutter, F. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.
- Monti, F., Boscaini, D., Masci, J., Rodola, E., Svoboda, J., and Bronstein, M. M. Geometric deep learning on graphs and manifolds using mixture model cnns. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 5115–5124, 2017.
- Pinkus, A. Approximation theory of the mlp model in neural networks. *Acta numerica*, 8:143–195, 1999.
- Qi, C. R., Su, H., Mo, K., and Guibas, L. J. Pointnet: Deep learning on point sets for 3d classification and segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 652–660, 2017.
- Ranade, R., Nabian, M. A., Tangsali, K., Kamenev, A., Hennigh, O., Cherukuri, R., and Choudhry, S. Domino: A decomposable multi-scale iterative neural operator for modeling large scale engineering simulations. *arXiv preprint arXiv:2501.13350*, 2025.
- Samareh, J. Aerodynamic shape optimization based on free-form deformation. In *10th AIAA/ISSMO multidisciplinary analysis and optimization conference*, pp. 4630, 2004.
- Su, J., Ahmed, M., Lu, Y., Pan, S., Bo, W., and Liu, Y. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.
- Tu, J., Yeoh, G. H., Liu, C., and Tao, Y. *Computational fluid dynamics: a practical approach*. Elsevier, 2023.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- Vatani, P., Elrefaie, M., Nazarpour, F., and Ahmed, F. TripOptimizer: Generative three-dimensional shape optimization and drag prediction using triplane variational autoencoder networks. *Physics of Fluids*, 37(12), 2025.
- Wu, H., Luo, H., Wang, H., Wang, J., and Long, M. Transolver: A fast transformer solver for PDEs on general geometries. In *International Conference on Machine Learning*, 2024.
- Yazdani, R. Steady and unsteady numerical analysis of the drivaer model. 2015.
- You, Y., Zhao, C., Zhang, H., Xu, M., and Fua, P. PhysGen: Physically grounded 3d shape generation for industrial design. *arXiv preprint arXiv:2512.00422*, 2025.
- Zhang, B. and Sennrich, R. Root mean square layer normalization. *Advances in neural information processing systems*, 32, 2019.
- Zhang, Z. Improved adam optimizer for deep neural networks. In *2018 IEEE/ACM 26th international symposium on quality of service (IWQoS)*, pp. 1–2. Ieee, 2018.
- Zhao, H., Jiang, L., Jia, J., Torr, P. H., and Koltun, V. Point transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 16259–16268, 2021.

Supplementary Material

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A. Details of Problem Formulation

This section explains the vehicle aerodynamic prediction and optimization problem that UniDrag addresses. We first describe why this problem is challenging, then explain our approach in an accessible way.

A.1. Why Vehicle Aerodynamics is Hard

When a car drives at highway speed, air resistance (drag) is the dominant force opposing motion. The drag coefficient C_d quantifies how "slippery" a vehicle is—lower C_d means better fuel efficiency. For electric vehicles, reducing C_d by just 0.01 can extend driving range by 5–10 km, which matters significantly for consumer adoption.

To predict C_d accurately, engineers use Computational Fluid Dynamics (CFD), which simulates airflow around the vehicle by solving physics equations on millions of computational cells. The problem is that high-fidelity CFD is extremely slow: a single simulation takes hours to weeks depending on accuracy requirements (Table 5). This means engineers can only test a handful of design variants during a multi-year development cycle.

A.2. The Gap Between Prediction and Design

Existing AI approaches fall into two categories, each with significant limitations:

Prediction-only methods (like GeoTransolver (Adams et al., 2025), AB-UPT (Alkin et al., 2025)) can predict airflow fields quickly, but they cannot tell engineers *how* to improve the design. Engineers must guess shape changes and verify them one by one—essentially trial-and-error with a faster simulator.

Generative methods (like PhysGen (You et al., 2025), TripOptimizer (Vatani et al., 2025)) attempt to directly generate improved shapes. However, they often produce invalid results: asymmetric vehicles, self-intersecting surfaces, missing wheels, or "bumpy" surfaces that exploit model

weaknesses rather than achieving genuine aerodynamic improvement.

UniDrag bridges this gap by combining accurate prediction with constrained optimization—it not only predicts airflow but also automatically finds shape improvements that are guaranteed to be manufacturable and symmetric.

A.3. What UniDrag Predicts

UniDrag takes a vehicle geometry (represented as a 3D surface mesh with 50,000 points) and predicts four types of outputs:

Surface pressure field: The pressure distribution on the vehicle surface, which determines where air pushes against the car. High pressure at the front and low pressure at the rear create drag.

Volume velocity field: The 3D airflow pattern around the vehicle (200,000 points), showing how air accelerates, decelerates, and forms wake structures behind the car.

Drag coefficient C_d : A single number summarizing total aerodynamic resistance. Typical passenger cars have C_d between 0.25–0.35.

Drag build-up profile: A 100-segment curve showing where drag accumulates along the vehicle length. This diagnostic reveals which regions (front fascia, A-pillar, rear wake) contribute most to total drag, directly guiding optimization efforts. The final segment of this profile equals C_d .

A.4. How UniDrag Optimizes Shapes

Given a baseline vehicle, UniDrag finds shape modifications that reduce C_d while satisfying engineering constraints. The key ideas are:

Free-Form Deformation (FFD): Instead of modifying individual surface points (which could create invalid geometry), we place a 3D grid of control points around the vehicle (Samareh, 2004). Moving these control points smoothly deforms the surface, guaranteeing that the result remains a valid, smooth shape suitable for manufacturing.

Bilateral symmetry: Cars must be left-right symmetric for aesthetics and handling. We only optimize control points on one side and automatically mirror them to the other.

Region freezing: Wheels, windows, and underbody are frozen—we only modify aerodynamically relevant regions (front, rear, roof) identified by the build-up profile.

Robustness via averaging: Neural networks can be "fooled" by adversarial inputs that appear good but fail in reality. To prevent this, we average predictions over multiple random point samples (Expectation-over-Transformation).

Table 5. Computational cost of CFD simulations at different fidelity levels.

Method	Resolution	Time	Compute Cost
Steady RANS	50–100M cells	4–12 hours	100–500 core-hours
Unsteady RANS	50–100M cells	1–3 days	1k–5k core-hours
Large Eddy Simulation	200M+ cells	2–4 weeks	50k–200k core-hours

If a shape change genuinely reduces drag, it will do so consistently; if it exploits model artifacts, the effect averages out.

A.5. Connection to Main Text

The components described above map directly to UniDrag’s architecture (Section 3 of the main text):

- Surface/volume prediction → ePSA backbone with 3D-RoPE (Section 3.2)
- Drag and build-up prediction → Drag Build-up Head with gradient isolation (Section 3.4)
- FFD-based optimization → Robust Differentiable Optimization (Section 3.6)
- Robustness via averaging → Expectation-over-Transformation (Section 3.6)

A.6. Comparison with Existing Approaches

Table 6 summarizes the key differences. UniDrag uniquely combines accurate multi-field prediction with constrained, robust optimization that produces CFD-verifiable results.

B. Detailed Model Architecture

This section provides additional architectural details of UniDrag, expanding upon the method description in the main text (Section 3).

B.1. Enhanced Physics-Sliced Attention (ePSA)

As described in Section 3.2, the backbone consists of 12 ePSA blocks. For each point i , we first aggregate neighborhood features within radius r using an MLP encoder $\psi(\cdot)$ (Pinkus, 1999) that processes relative positions and normals. The aggregated features are standardized to ensure scale-invariant encoding, and the augmented feature $\mathbf{h}_i^{\text{aug}} = \text{Concat}[\mathbf{h}_i, \mathbf{g}_i]$ is used for slice routing. We use $S = 32$ learnable physics slices, where each slice represents a physical structure prototype (e.g., boundary layer region, separation bubble, wake core). The routing uses a temperature-controlled softmax with learnable temperature initialized to $\tau = 0.5$. Each slice aggregates information from all points weighted by routing scores, and the weighted spatial centroid serves as the slice’s 3D position.

Before computing attention, we apply RMS normalization to query and key vectors to prevent score explosions from CFD’s large magnitude variations. The hidden dimension $d = 384$ is split into three equal parts for x, y, z axes, and 3D rotary transformations encode spatial relationships between slices. With $N \approx 50,000$ points and $S = 32$ slices, ePSA reduces attention complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(NS + S^2)$, yielding approximately 1500× reduction.

B.2. Gated Expert Routing (GER)

As described in Section 3.3, the last 4 ePSA blocks replace standard FFN with Mixture-of-Experts. We use $M = 3$ experts, each a two-layer MLP with GELU activation and $2\times$ expansion ratio ($384 \rightarrow 768 \rightarrow 384$). A learnable router computes soft weights over experts for each token via $\alpha_i = \text{softmax}(\mathbf{W}_g \mathbf{h}_i')$, and the output is a weighted combination of all expert outputs. To prevent expert collapse, we add an auxiliary loss encouraging balanced expert usage.

B.3. Modality-Protected Learning (MPL)

As described in Section 3.4, we design information flow to respect physical dependencies while preventing negative transfer. The surface head predicts surface pressure directly from surface features $\tilde{\mathbf{h}}^{(s)}$ via an MLP ($384 \rightarrow 256 \rightarrow 1$), receiving full gradient flow. The volume head predicts velocity from volume features with surface context passed through a stop-gradient operation, allowing volume predictions to benefit from surface information while preventing volume errors from corrupting surface representations. The drag build-up head predicts both C_d and the 100-segment build-up profile jointly, where C_d equals the final segment. It receives attention-pooled surface features with gradient isolation, geometry encoding, and global conditions (Reynolds number, freestream velocity). The stop-gradient on surface features prevents strong drag gradients from hijacking the backbone.

B.4. Robust Differentiable Optimization (RDO)

As described in Section 3.6, we freeze the trained surrogate and optimize geometry using FFD with robust constraints. A $5 \times 5 \times 5$ control lattice (125 control points) is placed around the vehicle, with surface points deformed via trivariate Bernstein polynomials guaranteeing C^2 continuous deformations. After bilateral symmetry reduction (optimizing only $y > 0$ and mirroring), we optimize 75 independent

Table 6. Comparison of problem formulations across different approaches.

Aspect	Prediction-Only	Generative	UniDrag (Ours)
Can predict airflow fields?	Yes	No	Yes
Can optimize shapes?	No (manual only)	Yes	Yes
Guarantees valid geometry?	N/A	No	Yes (FFD)
Enforces symmetry?	N/A	No	Yes
Robust to model artifacts?	N/A	No	Yes (EoT)
CFD-verifiable results?	N/A	Rarely	Always

parameters.

We target aerodynamically significant regions (front 15%, rear 15%, roof 20%) identified by the build-up profile, using a C^3 -continuous weight function that smoothly transitions from full deformation to zero at region boundaries. Wheels, windows, and underbody are frozen. Smoothness regularization penalizes non-smooth control point configurations through second and third-order differences and neighbor consistency terms. A displacement penalty enforces maximum deformation bounds (typically 3% of vehicle length). To prevent adversarial exploitation, we sample $M = 5$ random subsets of 50,000 surface points per optimization step and average predictions, smoothing the loss landscape and suppressing deformations that exploit specific point configurations.

Rationale for the region weighting $w(t) = [35t^4 - 84t^5 + 70t^6 - 20t^7]^2$. The specific form of $w(t)$ in Eq. (16) is *not* an empirical pick. It is the lowest-order univariate polynomial that satisfies the smoothness constraints imposed by the rest of the pipeline, and it admits a short derivation.

(i) *Constraint derivation.* We seek a transition function $s(t)$ on $[0, 1]$ with $s(0) = 0$, $s(1) = 1$, and $s^{(k)}(0) = s^{(k)}(1) = 0$ for $k = 1, 2, 3$. These eight linear conditions uniquely determine the seventh-order “smootheststep” $s(t) = 35t^4 - 84t^5 + 70t^6 - 20t^7$, which is exactly C^3 . Any lower-order polynomial cannot meet all eight conditions; any higher-order one would add unconstrained degrees of freedom without improving the continuity class.

(ii) *Why C^3 specifically.* The trivariate Bernstein FFD already produces C^2 deformations of the surface. Multiplying by a region weight $w(\cdot)$ preserves C^2 only if w is at least C^2 , so C^2 would be the minimum acceptable choice. We use C^3 to retain one extra derivative of smoothness across the region boundary, because automotive Class-A surfaces require curvature (G^2) continuity and CFD boundary-layer solvers are sensitive to *curvature-rate* discontinuities (jumps in s'''), which the C^2 alternative still allows.

(iii) *Why squaring.* Squaring does not raise the continuity class but tightens the boundary decay: near $t = 0$ we have $s(t) = O(t^4)$ and hence $w(t) = s(t)^2 = O(t^8)$, shrinking

the residual transition amplitude by roughly five orders of magnitude relative to the canonical cubic C^1 smoothstep $s_1(t) = 3t^2 - 2t^3$ ($O(t^2)$ near the boundary). This is what makes the deformed region visually seamless against the frozen region while still allowing full-magnitude editing in the interior.

(iv) *Comparison with simpler masks.* During development we tested three standard alternatives on the same FFD setup: a linear ramp (C^0), the cubic C^1 smoothstep $3t^2 - 2t^3$, and the quintic C^2 smootherstep $6t^5 - 15t^4 + 10t^3$. The linear ramp introduced a visible slope discontinuity at every region boundary and was rejected by our C^2 check on the deformed mesh. The cubic smoothstep produced occasional surface-pressure spikes in CFD verification near the front/roof junction, consistent with its C^1 -only smoothness. The quintic C^2 smootherstep eliminated those spikes but still caused λ_{C_3} to react at the boundary, because its s''' does not vanish there. Our C^3 squared form removes all three failure modes simultaneously, at no measurable runtime cost (a single $w(t)$ evaluation per surface point per step). We therefore view Eq. (16) as the minimal smooth mask consistent with FFD continuity, Class-A surface requirements, and the smoothness regularizer used elsewhere in the optimization, rather than as a tuned hyperparameter.

C. Additional Implementation Details

This section provides comprehensive implementation details for UniDrag, including model architecture specifications, training configurations, and optimization hyperparameters.

C.1. Model Architecture

UniDrag adopts a Transformer-based architecture with 12 enhanced Physics-Sliced Attention (ePSA) blocks and a hidden dimension of 384. Each block uses 8 attention heads and routes point tokens to 32 learnable physics slices. The last 4 blocks incorporate Gated Expert Routing (GER) with 3 experts per block, while the first 8 blocks use standard feed-forward networks. Local geometry features are extracted within a normalized radius of 0.05 using ball queries and MLP aggregation.

Hyperparameter	Value
Hidden dimension d	384
Number of ePSA blocks	12
Number of attention heads	8
Number of slices S	32
MLP expansion ratio	4
GER blocks (with MoE)	last 4 blocks
Number of experts M	3
Local geometry radius r	0.05 (normalized)
Surface tokens	50,000 per sample
Volume tokens	200,000 per sample
Surface head	MLP 384 $\rightarrow$ 256 $\rightarrow$ 1
Volume head	MLP 384 $\rightarrow$ 256 $\rightarrow$ 3
Drag build-up head	1D Conv, 100 bins

Table 7. UniDrag model architecture specifications.

The model processes approximately 50,000 surface points and 200,000 volume points per sample. The surface head is a two-layer MLP (384 $\rightarrow$ 256 $\rightarrow$ 1) predicting scalar pressure at each surface point. The volume head has the same structure but outputs 3D velocity vectors (384 $\rightarrow$ 256 $\rightarrow$ 3). The drag build-up head uses 1D convolutions to predict the 100-segment streamwise drag profile, where the final segment equals C_d . Table 7 summarizes the architecture specifications.

C.2. Training Configuration

UniDrag is trained end-to-end for 100 epochs using the AdamW optimizer with learning rate 10^{-3} and weight decay 10^{-6} . We use cosine annealing learning rate scheduling with 5 warmup epochs. Training is performed on 8 NVIDIA A100 GPUs (80GB each) with a total batch size of 8 and mixed FP16/FP32 precision for memory efficiency. Gradient clipping with max norm 1.0 ensures training stability.

All loss terms (surface pressure, volume velocity, C_d , and build-up profile) use equal weights of 1.0, which works well due to the relative error formulation that normalizes each loss to comparable scales. The GER auxiliary loss for load balancing uses a small weight of 0.01 to encourage balanced expert usage without dominating the main objectives. Table 8 summarizes the training hyperparameters.

C.3. FFD Optimization Configuration

Headline configuration. All main-paper results—including the 13.7% mean CFD-verified drag reduction, the 100% engineering-constraint satisfaction rate, the 21.3 mm average displacement, and every entry of the main quantitative tables—are produced with the single configuration summarized in Table 9: a global $5 \times 5 \times 5$ FFD lattice (125 control points, 75 free parameters after bilateral symmetry), Adam with step size 0.01 for 30 optimization steps, EoT multi-sampling with $M = 5$, regularization weights $\lambda_{C_2}/\lambda_{C_3}/\lambda_{nb}/\lambda_{disp} = 15/8/5/500$, and a maximum displacement of 3% of vehicle length. Any other grid resolution (e.g., $8 \times 6 \times 6$, $10 \times 8 \times 8$, $12 \times 10 \times 10$) or smooth-

Hyperparameter	Value
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	10^{-6}
LR schedule	cosine annealing
Warmup epochs	5
Total epochs	100
Batch size	8
Number of GPUs	$8 \times$ NVIDIA A100 (80GB)
Precision	mixed FP16/FP32
Gradient clipping	max norm 1.0
Surface loss weight λ_s	1.0
Volume loss weight λ_v	1.0
C_d loss weight λ_{C_d}	1.0
Build-up loss weight λ_b	1.0
GER auxiliary loss weight	0.01

Table 8. UniDrag training hyperparameters.

ness setting that appears later in this appendix (notably in Table 19 and the per-region visualizations) is used *exclusively* as an exploratory ablation and is *not* the configuration behind any headline number.

At deployment, we freeze the trained surrogate and optimize vehicle geometry using Free-Form Deformation (FFD). The FFD uses a $5 \times 5 \times 5$ control lattice with 125 control points. After enforcing bilateral symmetry (optimizing only $y > 0$ and mirroring), we have 75 independent parameters to optimize. We use the Adam optimizer with step size 0.01 for 30 optimization steps.

To prevent adversarial exploitation of surrogate fragility, we employ Expectation-over-Transformation (EoT) with $M = 5$ random point subsamples per step. Smoothness regularization includes C^2 and C^3 penalties on control point differences ($\lambda_{C_2} = 15$, $\lambda_{C_3} = 8$) and neighbor consistency ($\lambda_{nb} = 5$). A displacement penalty ($\lambda_{disp} = 500$) enforces maximum deformation of 3% vehicle length.

We target three aerodynamically significant regions identified by the build-up profile: front (0–15% streamwise), rear (85–100% streamwise), and roof (top 20% vertical). Wheels, windows, and underbody are frozen to preserve functional constraints. Table 9 summarizes the optimization hyperparameters.

D. Dataset Construction Details

This section provides comprehensive details on the construction of our large-scale vehicle aerodynamics dataset, covering geometry sources, standardization procedures, CFD simulation protocols, and quality assurance measures.

D.1. Overview and Design Objectives

The primary objectives of our dataset design are to ensure geometric diversity, physical fidelity, optimization support, and scale. Unlike parametric datasets (e.g., DrivAerNet (El-

Hyperparameter	Value
FFD grid resolution	$5 \times 5 \times 5$
Control points (after symmetry)	75
Optimizer	Adam
Step size	0.01
Number of steps	30
EoT samples M	5
C^2 smoothness λ_{C2}	15
C^3 smoothness λ_{C3}	8
Neighbor consistency λ_{nb}	5
Displacement penalty λ_{disp}	500
Max displacement d_{max}	3% of vehicle length
Front region	0–15% streamwise
Rear region	85–100% streamwise
Roof region	top 20% vertical
Frozen regions	wheels, windows, underbody

Table 9. FFD optimization hyperparameters.

refaie et al., 2024a)) that rely on morphing limited templates, our dataset combines licensed production-style models with Hunyuan 3D (Lai et al., 2025) to achieve greater topological diversity while strictly maintaining geometric validity suitable for CFD. Our dataset includes CFD-verified segment-wise C_d values over 100 partitions, which provides a principled signal to guide localized optimization.

D.2. Body Classes and Sample Distribution

Given the limited controllability of direct text-to-3D generation, we first use Hunyuanimage (Cao et al., 2025) to synthesize vehicle images from prompts with explicit model-type conditioning, and then generate high-fidelity 3D meshes from these images via an image-to-3D pipeline. We focus on common passenger-vehicle body classes, ensuring the dataset captures a wide range of aerodynamic characteristics:

- **Sedan (25.7%)**: Three-box configuration with a distinct trunk and low roofline, characterized by attached roof flow and a moderate wake.
- **SUV (26.7%)**: Tall two-box silhouette with an upright rear end, featuring a large frontal area and significant ground clearance effects.
- **MPV (22.6%)**: Compact two-box design with an integrated rear, exhibiting fastback-like flow and a smaller wake structure than sedans.
- **Sports car (25.0%)**: Extended roofline with a vertical rear and long wheelbase, similar to MPVs but with larger rear separation zones.

The total dataset comprises **15,000 standardized vehicle meshes**. The image and 3D mesh results are shown in the Fig. 9. The distribution is intentionally kept nearly uniform (difference $< 13\%$ between classes) to prevent the



Figure 9. High-fidelity images generated by Hunyuan (left) and the corresponding meshes (right).

surrogate model from overfitting to any single body type. To empirically validate the geometric diversity discussed above, Fig. 10 presents a qualitative gallery of randomly sampled geometries from the dataset. The samples are organized as a matrix where rows correspond to distinct body classes and columns represent individual instances after the standardization pipeline. Visual inspection reveals substantial *intra-class morphological variance*. Even within a single category, we observe significant heterogeneity in critical aerodynamic surfaces, including the rake of the windshield, roofline curvature, rear deck proportions, and localized styling cues. This structural diversity is a deliberate design choice: unlike parametric models that may produce repetitive, self-similar shapes, our dataset covers a continuous manifold of realistic vehicle topologies. This property is fundamental for the UniDrag framework, ensuring that the learned surrogate generalizes to novel geometries rather than overfitting to a limited set of canonical templates.

D.3. Geometry Sources

We construct the dataset from two complementary sources to balance physical realism with stylistic exploration:

1. **Licensed Production-Style Models**: We acquired several thousand high-fidelity meshes from licensed CAD repositories and automotive design partners. These models serve as the "anchors" of our dataset, grounding it in physical plausibility. They closely match real production vehicles in critical engineering proportions, wheelbase ratios, and detailed underbody layouts (including realistic ground clearance and wheel geometry), ensuring the simulation captures valid flow physics.
2. **Hunyuan3D-Generated 3D Assets**: To expand the topological diversity beyond existing production catalogs, we employed text- and image-conditioned 3D generation pipelines by using Hunyuanimage and Hunyuan 3D (Cao et al., 2025; Lai et al., 2025). Prompts were specifically engineered to target the four body classes while varying local styling details and proportions. Crucially, these generated concepts undergo the same rigorous standardization and quality checks as the licensed models; only geometries passing all val-

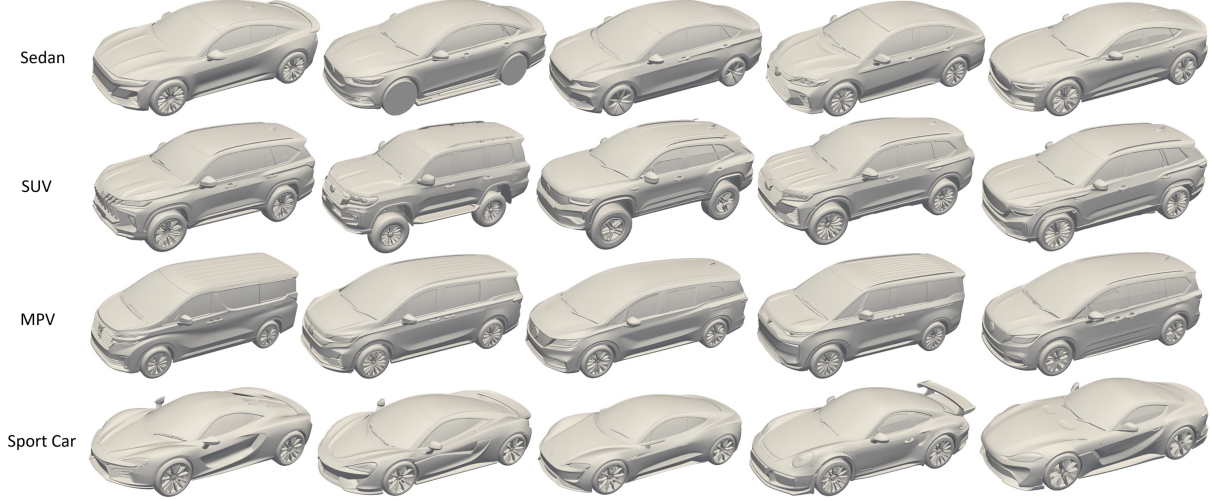


Figure 10. Qualitative overview of the UniDrag dataset geometry. The matrix displays randomly sampled geometry from four representative categories: Sedan, SUV, MPV, and Sports Car. Rows correspond to body classes, while columns present individual instances following our alignment and scale normalization pipeline. Observe the significant *intra-class morphological variance*—particularly in roof curvature, rear overhangs, and greenhouse proportions. This diversity underscores that the dataset captures a dense, realistic manifold of automotive shapes rather than limited template variations.

idation filters are retained. This approach allows the dataset to capture both current industry standards and creative variations representing potential future styling trends.

D.4. Dimension Priors and Normalization

To ensure physical realism, we enforce class-specific dimension ranges derived from automotive industry surveys. We define rigorous bounding boxes for each category: **Sedans** are constrained to lengths of 3.2–5.3 m, widths of 1.5–1.9 m, and heights of 1.5–1.7 m. **SUVs** feature larger envelopes with lengths up to 4.3–5.2 m and heights ranging from 1.8–2.0 m. Similarly distinct priors are applied to **MPVs** (L : 4.6–5.9 m) and **Sports cars** (L : 4.0–5.0 m). These intervals are slightly widened relative to strict production specifications to accommodate design variations while preserving physical plausibility.

Every raw mesh passes through a unified **Standardization Pipeline** to ensure topological and semantic consistency, as visualized in Fig. 11.

1. **Coordinate System Recovery:** We first perform PCA on the outer body shell to identify principal axes. Crucially, we utilize the geometric centroids of the wheels to disambiguate the orientation, fixing the nose direction ($-x$) and the vertical axis ($+z$) to establish a canonical frame.
2. **Ground Plane Alignment:** A ground plane is fitted through the lowest contact points of the four wheels. The vehicle is then translated such that its origin lies on

the ground plane directly beneath the geometric center, ensuring a consistent reference for CFD simulation.

3. **Class-Specific Size Projection:** Raw geometries are projected onto the valid class ranges. We employ a scaling strategy with a priority hierarchy of $L \rightarrow H \rightarrow W$, minimizing distortion while enforcing constraints. Scale factors are stored in metadata to allow for original dimension recovery.
4. **Mesh Health Validation:** The pipeline automatically detects and repairs topological defects. This includes fixing flipped normals, filling small holes, and resolving non-manifold edges to ensure the mesh is watertight (genus-0) and suitable for volume meshing.
5. **Region Tagging & Packaging:** Finally, surface triangles are semantically segmented into *body*, *underbody*, and *wheel* groups. This tagging enables the application of differential CFD boundary conditions (e.g., rotating walls for wheels). The final output is packaged as an STL file with accompanying JSON metadata containing the bounding box and body-class label.

D.5. Lattice-Boltzmann CFD Simulations

We leverage the Lattice-Boltzmann Method (LBM) (Chen & Doolen, 1998) to generate high-fidelity aerodynamic labels. Compared to steady-state RANS approaches, our LBM pipeline naturally captures transient flow structures and massive separation regions inherent in automotive aerodynamics, ensuring the surrogate model is trained on physics-compliant data with realistic flow unsteadiness.

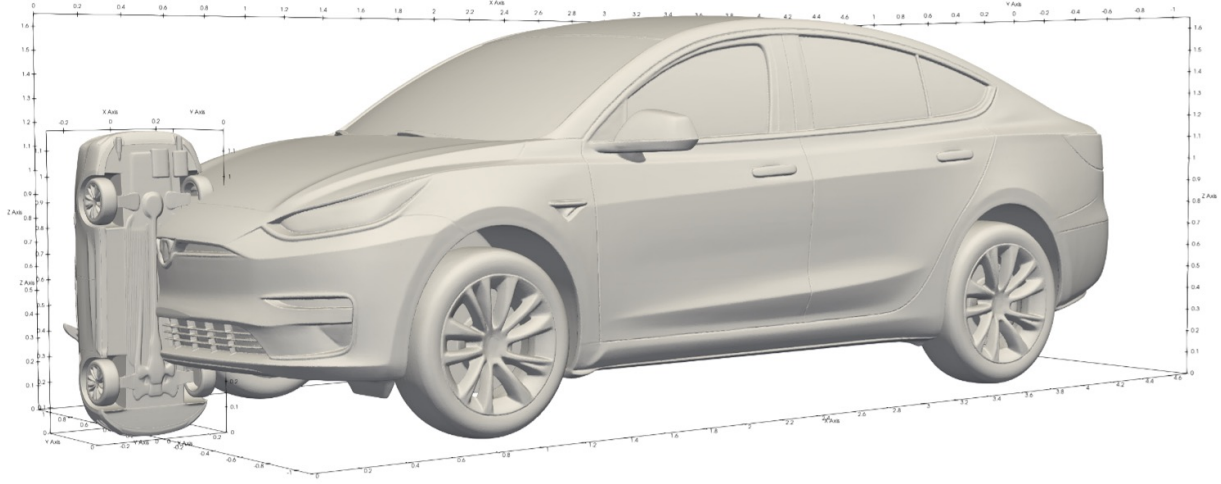


Figure 11. Visual demonstration of the geometric standardization pipeline. The figure illustrates the transformation of a raw 3D asset (depicted in the small inset with arbitrary orientation) into a simulation-ready geometry (main view). Raw inputs typically suffer from inconsistent scaling and random coordinate alignments. Our pipeline rectifies these discrepancies by registering the mesh to a unified canonical frame ($-x$ aligned with flow direction), enforcing strict ground-plane contact, and constraining the dimensions to the physical priors. This rigorous spatial normalization ensures that all samples share a consistent reference system, which is a prerequisite for learning robust aerodynamic features.

D.5.1. DIGITAL WIND TUNNEL CONFIGURATION

The simulation environment is constructed as a virtual “rolling-road” wind tunnel. To ensure consistency across varying vehicle lengths, we define a fixed global coordinate system aligned with the standardized vehicle frame.

Computational Domain: The domain is a rectangular box extending from $x_{\min} = -25$ m (upstream) to $x_{\max} = 50$ m (downstream) relative to the vehicle center. Laterally, the domain spans $y \in [-25, 25]$ m, and vertically it extends from the ground ($z = 0$) to $z_{\max} = 30$ m. This spatial extent is sufficiently large to minimize blockage effects and boundary interference.

Flow Conditions and Boundaries: We simulate a highway cruising scenario with a freestream velocity $U_{\infty} = 33$ m/s (≈ 120 km/h), air density $\rho = 1.184$ kg/m³, and kinematic viscosity $\nu = 1.51 \times 10^{-5}$ m²/s. The boundary conditions are configured to replicate physical wind tunnel tests:

- **Inlet & Outlet:** A uniform velocity profile is imposed at the inlet, while the outlet utilizes a pressure-based condition.
- **Moving Ground:** The ground plane is modeled as a *moving wall* translating at U_{∞} .
- **Rotating Wheels:** Wheel geometries are explicitly tagged and assigned a *rotating wall* boundary condition with angular velocity matched to the freestream speed. This setup prevents unrealistic boundary layer growth on the floor and correctly reproduces tire contact patch

physics.

- **Sides & Top:** Top and lateral boundaries are treated as slip walls.

D.5.2. MESH REFINEMENT STRATEGY

Spatial discretization is managed via a hierarchical block-structured Cartesian grid. As visualized in the upper panel of Fig. 12, this topology divides the computational volume into nested blocks, allowing for localized resolution scaling without disrupting the global data structure. We then employ a refinement strategy with up to nine nested levels to efficiently resolve multiscale flow features.

Grid Hierarchy: Refinement boxes are centered on the vehicle, with resolution increasing progressively from the far-field towards the body.

Finest Resolution: The finest grid level ($\Delta x_{\min} = 4$ mm) is applied to the vehicle surface and complex interaction zones (e.g., wheel spokes/rims) to capture geometric details.

Wake Resolution: Intermediate refinement levels cover the underbody and the near-wake region (extending approximately one vehicle length downstream) to resolve shear layers and flow separation.

D.5.3. SOLVER DYNAMICS AND TEMPORAL SETTINGS

The solver utilizes a robust Lattice-Boltzmann formulation with enhanced hybrid regularized collision operator. This

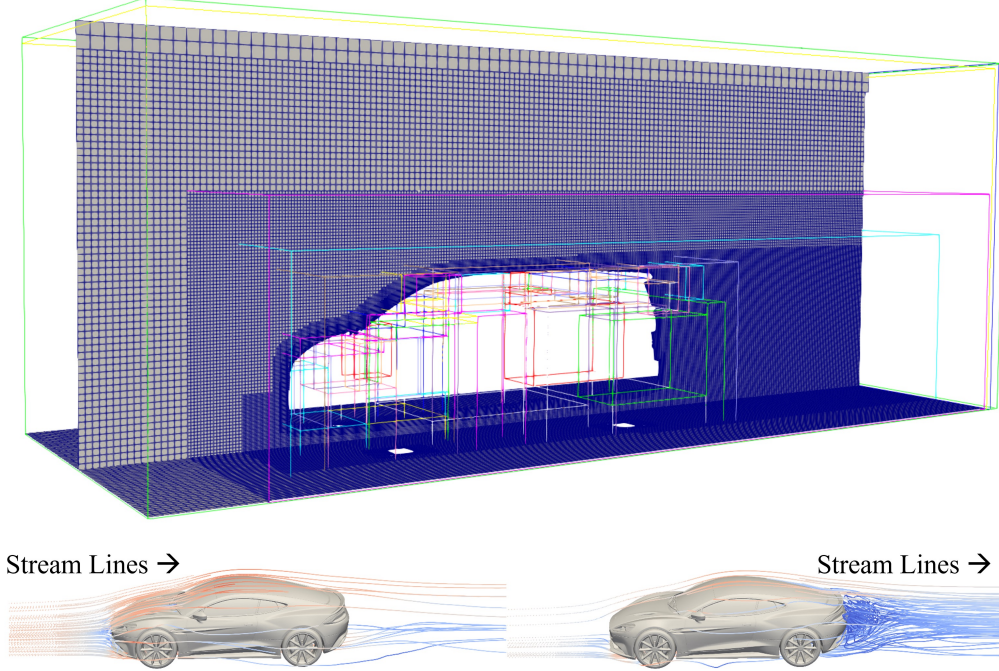


Figure 12. Computational domain discretization and flow visualization. **Upper panel:** Illustration of the hierarchical spatial discretization strategy. The domain is decomposed into nested Cartesian blocks (delineated by colored wireframe outlines), enabling efficient adaptive refinement around the vehicle geometry. **Lower panels:** Volumetric flow visualization using Lagrangian streamlines mapped with velocity magnitude. The traces highlight critical aerodynamic phenomena captured by the LBM solver, including the stagnation point at the front fascia (left) and the turbulent recirculation zones in the wake (right).

approach provides exceptional numerical stability for high-Reynolds-number flows and effectively resolves the time-dependent evolution of the velocity field.

Simulation Protocol: Each simulation is time-marched for a total physical duration of $T_{\text{sim}} = 3.0$ s.

1. **Initialization:** The initial phase allows for the development of the flow field and the flushing of startup transients.
2. **Statistical Averaging:** Aerodynamic statistics (including drag coefficient C_d and surface pressure) are averaged over the stable window $t \in [2.0, 3.0]$ s.
3. **Convergence:** This window ensures that the reported mean quantities are statistically converged, capturing the time-averaged effect of the unsteady wake.

To facilitate reproducibility and provide a quick reference for the simulation setup, Table 10 consolidates the rigorous physical and numerical configurations utilized throughout the dataset generation process.

D.6. Output Fields and Labels

For each simulated sample, we extract a multi-modal set of aerodynamic labels, capturing flow features from surface

integrals to volumetric distributions:

Surface Fields: Data is sampled at $N_s \approx 50,000$ vertices on the vehicle mesh $\mathcal{S}$. The feature set includes the dimensionless **Surface Pressure** (p_s), the **Wall Shear Stress** magnitude ($|\tau_w|$, in Pa), and the **Surface Velocity** vector ($\mathbf{u}_s$). These fields provide high-resolution supervision for boundary layer behavior.

Volume Fields: We query the flow field at $N_v \approx 200,000$ points within the domain $\mathcal{V}$. Crucially, these points are not uniformly distributed; they are *importance-sampled* with higher density near the vehicle surface and in the wake region to resolve high-gradient flow structures. Features include **Pressure** (p , in Pa) and the full **Velocity Vector** ($\mathbf{u} = (u, v, w)$, in m/s).

Integral Quantities: Global performance metrics are derived via surface integration. We first compute the mean segment-wise C_d values over the 100 partitions. The **Total Drag Coefficient** is computed as $C_d = \frac{2F_d}{\rho U_\infty^2 A_{\text{ref}}}$, where F_d is the integrated streamwise force (pressure + shear) and A_{ref} is the frontal area projected onto the yz -plane.

Build-up Profile $b(x)$: To allow for localized drag diagnosis, we compute a 1D streamwise profile $b(x)$. This is

Table 10. Comprehensive specification of the digital wind tunnel configuration. These parameters are tuned to ensure consistent, high-fidelity aerodynamic labeling across all 15,000 samples.

Configuration Parameter	Specification / Value
<i>Physical Conditions</i>	
Inlet Velocity (U_∞)	33 m/s (≈ 120 km/h)
Fluid Density (ρ)	1.184 kg/m ³
Kinematic Viscosity (ν)	1.51×10^{-5} m ² /s
Thermodynamic Pressure (p_{ref})	101,325 Pa
<i>Computational Domain & Grid</i>	
Streamwise Extent (x)	$[-25, +50]$ m
Lateral Span (y)	$[-25, +25]$ m
Vertical Height (z)	$[0, +30]$ m
Grid Refinement Strategy	9 hierarchical levels
Finest Cell Resolution	4 mm (near-body)
<i>Boundary & Temporal Settings</i>	
Ground Boundary	Moving wall ($v = U_\infty$)
Wheel Boundary	Rotating wall ($\omega = U_\infty/r_{\text{tire}}$)
Total Physical Time	3.0 s
Statistics Window	Last 1.0 s (Converged)

obtained by integrating the streamwise component of surface forces into $K = 100$ equal-width bins along the vehicle length, identifying exactly where drag is generated or accumulated.

D.7. Quality Assurance and Validation

The dataset generation was a computationally intensive campaign, consuming approximately 6×10^4 GPU-hours on a cluster of NVIDIA RTX 4090s. To guarantee the reliability of the resulting labels, we enforced a rigorous, multi-stage validation protocol.

Simulation Convergence and Physical Checks. Every simulation is continuously monitored. We enforce strict stability criteria: the drag coefficient variation must remain below 0.5% during the final averaging window ($t \in [2.0, 3.0]$ s), and the mass conservation residual must be less than 10^{-5} . Furthermore, we implement heuristic sanity checks on the output C_d ; samples falling outside physically plausible ranges (e.g., $C_d < 0.15$ or $C_d > 0.60$ for passenger cars) are flagged for manual topological inspection.

Validation against Experimental Benchmarks. To certify the absolute accuracy of our digital wind tunnel, we benchmarked the LBM solver against full-scale wind tunnel measurements of the DrivAer generic car model (Yazdani, 2015). We reproduced a set of canonical configurations (fastback/notchback/estateback) with matching Reynolds numbers and moving-ground boundary conditions.

Crucially, our solver demonstrates high fidelity without case-specific tuning. Across the tested configurations, the computed drag coefficients deviate from the experimental values

(Exp. PVT) by less than 3% (a maximum difference of typically < 0.01 or 10 drag counts). This strong quantitative agreement confirms that the solver correctly captures the sensitivity of drag to geometric changes (such as rear-end shape and underbody roughness), legitimizing its use as a ground-truth generator for the UniDrag dataset.

D.8. Qualitative Flow Topology Visualization

To intuitively assess the richness of the dataset, Figure 13 visualizes the resolved flow physics for distinct vehicle archetypes. The figure is organized as a matrix where rows represent different geometries (from top to bottom: an SUV, a fastback coupé, a low sports car, and an MPV) and columns display three fundamental aerodynamic fields:

1. **Surface Pressure (p_s):** clearly captures the high-pressure stagnation zones on the front fascia and the characteristic low-pressure suction peaks over the hood-windshield transition and roofline.
2. **Velocity Magnitude ($|\mathbf{u}|$):** reveals the massive flow acceleration over the greenhouse and the formation of low-velocity wake deficits behind the vehicle base.
3. **Lagrangian Streamlines:** visualize the complex 3D separation patterns. We observe that the boxy MPV (bottom row) generates a large, chaotic recirculation bubble, whereas the streamlined Wagon (top row) maintains attached flow over a longer portion of the roof, resulting in a narrower wake.

These detailed flow structures confirm that our LBM setup resolves critical transient features—such as A-pillar vortices and base separation—that are essential for training physics-aware surrogates.

D.9. Aerodynamic Performance Statistics

Figure 14 summarizes the distribution of the Total Drag Coefficient (C_d) across the four primary body classes in the dataset. The statistics are derived from the complete set of valid simulations.

The observed distributions align with physical expectations while highlighting specific characteristics of our data source:

- **Sedan:** Exhibits the lowest aerodynamic drag with a median of 0.278. The compact interquartile range indicates a consistent design language focused on efficiency.
- **MPV & SUV:** These classes show systematically higher drag (Medians of 0.307 and 0.337, respectively). This is physically consistent with their bluff rear-ends and larger wake cross-sections.

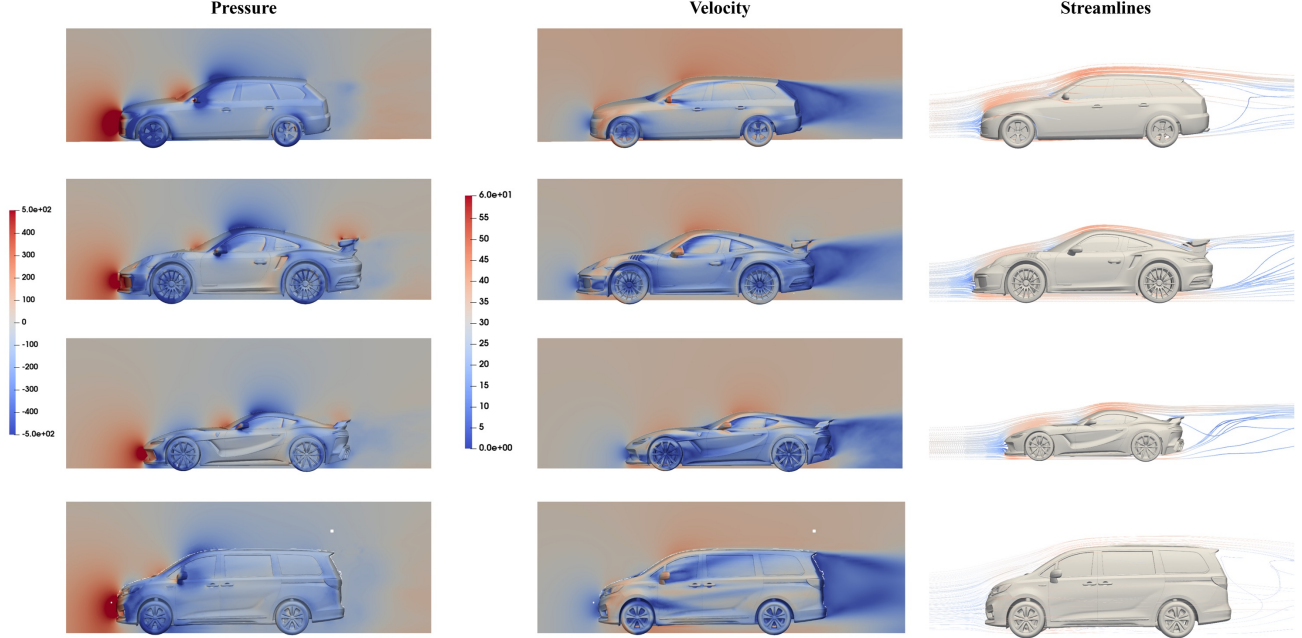


Figure 13. Visualization of multi-modal aerodynamic fields. We present LBM simulation results for four representative samples (Top to bottom: SUV, fastback coupé, sports car, MPV). Columns from left to right display: (1) Surface pressure distribution, highlighting stagnation and suction zones; (2) Volumetric velocity magnitude on the center-plane; and (3) 3D Streamlines showing wake topology. The solver captures distinct flow characteristics driven by geometry, from the attached flow on the streamlined coupes to the massive separation behind the bluff-body MPV.

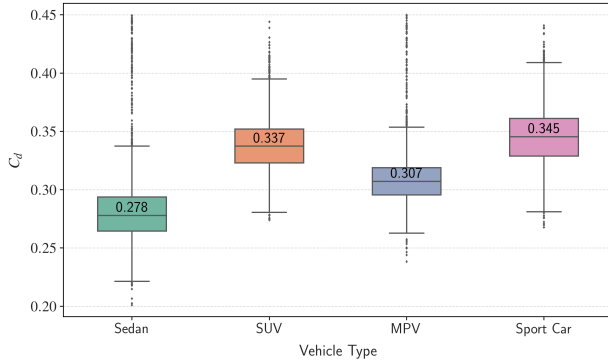


Figure 14. Distribution of Total Drag Coefficient (C_d) by body class. The boxplot illustrates the aerodynamic diversity within our dataset. **Sedans** are the most efficient class (Median $C_d \approx 0.278$). **MPVs** and **SUVs** show higher drag (Medians ≈ 0.307 and 0.337) due to larger frontal separation. **Sports Cars** in this dataset exhibit a relatively high median ($C_d \approx 0.345$), reflecting a bias towards high-downforce geometries with aggressive aero-features.

- **Sport Car:** Interestingly, this category displays a high median drag of 0.345 with significant variance. While sports cars have low frontal areas, the models in our dataset often feature aggressive aerodynamic add-ons (wings, splitters) and wider tires designed for downforce rather than pure drag reduction, leading to a higher induced drag penalty compared to clean-sheet sedans.

D.10. Comparison with Existing Datasets

Table 11 contextualizes UniDrag within the field. By combining LBM fidelity (Chen & Doolen, 1998), large scale, and multi-field labels, UniDrag uniquely supports both high-accuracy prediction and physics-based optimization.

Table 11. Comparison with existing vehicle aerodynamic datasets. Prior datasets are limited in scale, vehicle diversity, and lack optimization-ready annotations (e.g., drag build-up profiles). Our dataset is the first to provide comprehensive multi-field outputs with explicit support for differentiable shape optimization.

Dataset	Size	Vehicle Classes	CFD Type	Optimization-Ready
DrivAerNet	4k	1 (Sedan only)	RANS	No
DrivAerNet++	8k	3	RANS	No
Ours (UniDrag)	15k	4	LBM	Yes

E. Traditional Optimizer Method

Conventional generative shape-optimization pipelines (Vatani et al., 2025; You et al., 2025) are tightly coupled to the chosen generator and its learned manifold, making optimization outcomes highly sensitive to generation fidelity and dataset bias. Latent- or trajectory-based updates further offer limited locality and interpretability, and can drift into poorly supported regions without explicit physical/field-level constraints. As a result, when informative training samples are scarce and the design space is only partially covered, such optimization tends to be brittle and exhibits weak out-of-distribution generalization and robustness in practice.

E.1. TripOptimizer

TripOptimizer (Vatani et al., 2025) targets the high computational cost of CFD-based aerodynamic shape optimization by learning a differentiable generative surrogate that (i) reconstructs high-fidelity vehicle geometry from point clouds and (ii) predicts the drag coefficient C_d in a shared latent space, enabling gradient-based performance-driven shape refinement with substantially fewer high-fidelity CFD calls.

Principle. TripOptimizer is a VAE whose decoder uses a triplane-based implicit representation to reconstruct geometry and a prediction head to regress C_d . The training objective is a weighted sum of reconstruction, drag prediction, KL regularization, and optional geometric regularizers:

$$\mathcal{L}_{\text{total}} = w_{\text{recon}}\mathcal{L}_{\text{recon}} + w_{C_d}\mathcal{L}_{C_d} + w_{\text{KL}}\mathcal{L}_{\text{KL}} + w_{\text{bdry}}\mathcal{L}_{\text{bdry}} + w_{\text{tri}}\mathcal{L}_{\text{tri}}. \quad (20)$$

Its C_d head uses self-attention of the form

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V. \quad (21)$$

For optimization, instead of directly updating the latent code z , TripOptimizer refines a subset of encoder parameters ϕ' to alter how a fixed input point cloud X_{init} is mapped into latent space, while keeping the decoder and C_d head frozen:

$$\mathcal{L}_{\text{opt}}(\phi') = \left(\text{Head}_{C_d}(\mu_{\phi'}(X_{\text{init}}); \theta_{C_d}) - C_d^{\text{target}}\right)^2. \quad (22)$$

Implementation. The pipeline takes a vehicle point cloud as input, encodes it into a latent representation, decodes triplane features to a semi-continuous occupancy field, and extracts a mesh (e.g., marching cubes + smoothing). During optimization, only later layers of the encoder are set trainable and updated by backpropagating $\nabla_{\phi'}\mathcal{L}_{\text{opt}}$; the input point cloud remains unchanged, which constrains edits to remain on/near the learned manifold.

Limitations. (i) The method depends on the coverage and bias of the training dataset (e.g., template-/morph-based

vehicles), which can limit out-of-distribution generalization. (ii) The objective is primarily driven by a global scalar C_d , providing limited direct supervision for localized flow/pressure phenomena unless additional field predictors or constraints are introduced. (iii) Optimizing encoder parameters may be less interpretable and can be sensitive to which layers are unfrozen and to step-size schedules.

E.2. PhysGen

PhysGen (You et al., 2025) addresses the instability of physics-guided shape updates that may drift off the shape manifold by integrating (i) a physics-aware latent representation and (ii) an alternating generation paradigm that couples manifold-preserving flow-matching updates with physics-based refinement, aiming to generate shapes that are both geometrically plausible and physically favorable.

Principle. PhysGen first trains a Shape-and-Physics VAE (SP-VAE) that jointly models geometry and aerodynamics in a shared latent space. Shape is represented as an SDF, and physics includes surface pressure and drag. The staged training uses:

$$\mathcal{L}_{\text{shape}} = \lambda_{\text{sdf}}\|s - \hat{s}\|_2^2 + \lambda_{\text{KL}}\mathcal{L}_{\text{KL}}, \quad (23)$$

$$\mathcal{L}_{\text{press}} = \|p - \hat{p}\|_1 + \|p - \hat{p}\|_2^2, \quad (24)$$

$$\mathcal{L}_{\text{drag}} = \|C_d - \hat{C}_d\|_1 + \|C_d - \hat{C}_d\|_2^2, \quad (25)$$

$$\mathcal{L}_{\text{total}} = \lambda_s \mathcal{L}_{\text{shape}} + \lambda_p \mathcal{L}_{\text{press}} + \lambda_d \mathcal{L}_{\text{drag}}. \quad (26)$$

For generation, PhysGen uses rectified-flow matching in latent space. The noisy interpolation and target velocity are:

$$z_{t_n} = t_n z_1 + (1 - t_n)\epsilon, \quad \epsilon \sim \mathcal{N}(0, 1), \quad (27)$$

$$u_{t_n} = \frac{dz_{t_n}}{dt_n} = z_1 - \epsilon, \quad (28)$$

and the reverse update with the learned velocity $\hat{u}$ is:

$$z'_{t_{n+1}} = z_{t_n} - (t_{n+1} - t_n)\hat{u}(z_{t_n}, t_n, c). \quad (29)$$

Physics guidance is injected via the drag decoder D_d using a gradient step toward a target drag d_{tar} :

$$z_{t_{n+1}} = z'_{t_{n+1}} - \lambda_d \nabla_{z_{t_n}} \frac{1}{2} \|D_d(z_{t_n}) - d_{\text{tar}}\|_2^2. \quad (30)$$

Then, a physical refinement stage uses the decoded surface pressure to compute directional forces

$$F_s = \sum_{i=1}^V p_i n_{s,i} A_i, \quad s \in \{x, y, z\}, \quad (31)$$

and optimizes a force-based objective (e.g., drag minimization and stability constraints):

$$\mathcal{L} = \lambda_x \|F_x\|_2 + \lambda_y \|F_y\|_2 + \lambda_z \text{ReLU}(F_z). \quad (32)$$

Hyperparameter	Value
Architecture	Triplane-VAE + C_d regressor
Latent dimension	128
Optimiser	AdamW, lr 5×10^{-5} , wd 10^{-6}
Training epochs	80
LR schedule	StepLR (step 5, $\gamma=0.75$)
Loss function	SmoothL1, $\beta=0.05$
Loss weights	$w_{\text{recon}}=1$, $w_{C_d}=10$, $w_{\text{bdry}}=1$, $w_{\text{ini}}=5 \times 10^{-4}$
Optimization strategy	fine-tune last 3 encoder layers
Optimization lr	10^{-5}
Optimization steps	100
CFD verification	OpenFOAM simpleFoam + SST
Inlet velocity	30 m/s
CFD averaging	last 500 iterations

Table 12. **Implementation details of the TripOptimizer baseline.** TripOptimizer (Vatani et al., 2025) uses a Triplane-VAE with a drag regressor, optimizing in latent space via encoder fine-tuning.

Hyperparameter	Value
Architecture	SP-VAE (SDF + pressure/ C_d decoders) + DiT
Training epochs	1000/1500/1500 (shape/press/drag) + 500 joint
Loss weights	$\lambda_{\text{shape}}=10$, $\lambda_{\text{physics}}=0.1$, $\lambda_{\text{drag}}=10$
Flow sampling steps	100
Guidance weight λ_d	0.03
Refinement iterations K	20 (25 velocity + 20 refine per iter)
Noise schedule $t_{n.s}$	0.75
Force weights	$\lambda_x=0.2$, $\lambda_y=0.1$, $\lambda_z=0.1$
CFD verification	OpenFOAM simpleFoam + SST
Inlet velocity	30 m/s
CFD iterations	2500 (avg last 500)

Table 13. **Implementation details of the PhysGen baseline.** PhysGen (You et al., 2025) uses a physics-guided diffusion model with alternating flow updates and force-based refinement.

An alternating strategy repeats flow updates and physical refinement to balance manifold fidelity and physics optimality.

Implementation. SP-VAE consists of an encoder and multiple decoders (shape/SDF, pressure field, and scalar drag). During sampling, the method runs N flow-matching steps with physics-regularized updates (Eq. 30), followed by M refinement steps that backpropagate $\nabla_z \mathcal{L}$ (Eq. 32) through the decoders into the latent code, and iterates this alternating procedure for stable convergence.

Limitations. (i) Guidance quality is bounded by the accuracy/calibration of learned physics decoders (D_d , pressure decoder); model bias can misguide gradients. (ii) The alternating scheme introduces additional hyperparameters (e.g., N , M , λ 's) and can be sensitive to step sizes. (iii) The physics objectives are task-specific (here, aerodynamics/forces); adding manufacturability, constraints, or multi-physics typically requires extra supervision or constraint handling.

F. Evaluation Metrics and Baseline Implementations

This section provides detailed definitions of evaluation metrics and implementation details for baseline methods.

F.1. Evaluation Metrics

We evaluate UniDrag across two dimensions: prediction accuracy and optimization effectiveness.

F.1.1. PREDICTION METRICS

Coefficient of Determination (R^2). For scalar predictions including drag coefficient C_d and the 100-segment build-up profile, we use:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (33)$$

where y_i is the ground truth, $\hat{y}_i$ is the prediction, and $\bar{y}$ is the mean of ground truth values. $R^2 = 1$ indicates perfect prediction; higher is better.

Relative L_2 Error. For field predictions (surface pressure, volume velocity), we use:

$$\text{Rel-}L_2 = \frac{\|\hat{f} - f\|_2}{\|f\|_2} \quad (34)$$

where f and $\hat{f}$ denote the ground truth and predicted fields, respectively. This normalization ensures comparability across fields with different magnitudes; lower is better.

Relative L_1 Error. We also report relative L_1 error, which is more robust to outliers:

$$\text{Rel-}L_1 = \frac{\|\hat{f} - f\|_1}{\|f\|_1} \quad (35)$$

Build-up MAE. For the 100-segment drag build-up profile, we report mean absolute error:

$$\text{MAE} = \frac{1}{100} \sum_{k=1}^{100} |b_k - \hat{b}_k| \quad (36)$$

where b_k and $\hat{b}_k$ are the ground truth and predicted drag contributions at segment k .

F.1.2. OPTIMIZATION METRICS

CFD-Verified Drag Reduction. The primary optimization metric measures actual aerodynamic improvement:

$$\Delta C_d^{\text{CFD}} = \frac{C_d^{\text{opt}} - C_d^{\text{orig}}}{C_d^{\text{orig}}} \times 100\% \quad (37)$$

where C_d^{orig} and C_d^{opt} are CFD-computed drag coefficients before and after optimization. Negative values indicate drag reduction.

Success Rate. Percentage of test samples achieving genuine C_d reduction after CFD verification:

$$\text{Success Rate} = \frac{|\{i : C_{d,i}^{\text{opt}} < C_{d,i}^{\text{orig}}\}|}{N_{\text{test}}} \times 100\% \quad (38)$$

Surface Displacement. Average vertex displacement quantifies the magnitude of shape modification:

$$\bar{d} = \frac{1}{N_s} \sum_{i=1}^{N_s} \|\mathbf{x}_i^{\text{opt}} - \mathbf{x}_i^{\text{orig}}\|_2 \quad (39)$$

We report this both in millimeters and as a percentage of vehicle length to assess manufacturability.

F.2. Baseline Implementations

This section provides implementation details for prediction baselines (Transolver, GeoTransolver, AB-UPT) and optimization baselines (TripOptimizer and PhysGen, described in Section E).

F.3. Transolver baseline implementation

We adopt Transolver (Wu et al., 2024), a Transformer-based surrogate designed for large-scale irregular CFD discretizations. Each mesh node’s spatial coordinates (x, y, z) are lifted to a C -dimensional feature space through a two-layer MLP, with a learnable global offset added for expressivity. Transolver replaces full self-attention with a physics-informed slice-deslice mechanism: node features are softly aggregated into G latent slice tokens, self-attention is performed in this reduced space, and updated features are mapped back to nodes. This reduces complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NGC + G^2C)$. Each layer follows a standard pre-normalized Transformer (Vaswani et al., 2017) design with slice-based attention and feed-forward network. To ensure fair comparison, we scale up Transolver to match UniDrag’s capacity. The hyperparameters are summarised in Table 14.

F.4. GeoTransolver baseline implementation

GeoTransolver (Adams et al., 2025) is a multiscale geometry-aware physics attention transformer for high-fidelity CAE surrogate modeling. The model takes three input modalities: local state slices with spatial coordinates and field features, geometry data with coordinates and auxiliary attributes, and global parameters (e.g., Reynolds number). Multi-scale ball queries extract neighborhood features at multiple radii, which are processed via MLPs and concatenated with local inputs. Global geometry encodings are

Hyperparameter	Value
Input features per node	$(x, y, z, \mathbf{n}) \in \mathbb{R}^6$
Hidden dimension C	384
Number of blocks L	12
Number of heads H	8
Number of slices G	32
MLP expansion ratio r	4
Preprocessing MLP	$6 \rightarrow 768 \rightarrow 384$, GELU
Block MLP	$384 \rightarrow 1536 \rightarrow 384$, GELU
Temperature initialisation	0.5 (learnable)
Optimiser	AdamW, lr 10^{-3} , wd 10^{-6}
LR schedule	cosine annealing
Loss	relative L_2 on surface/volume fields
Precision	mixed FP16/FP32

Table 14. Implementation details of the Transolver baseline. We scale up the original Transolver (Wu et al., 2024) to match UniDrag’s model capacity, using 12 blocks with hidden dimension 384 and 32 physics slices.

Hyperparameter	Value
Hidden dimension d	384
Number of GALE blocks	12
Number of attention heads	8
Number of slices G	32
MLP expansion ratio	4
Multi-scale ball query radii	$\{0.02, 0.05, 0.1\}$
Surface tokens	50,000 per sample
Volume tokens	200,000 per sample
Geometry context aggregation	mean + max pooling
Cross-attention gate	sigmoid, learnable
Optimiser	AdamW, lr 10^{-3} , wd 10^{-6}
LR schedule	cosine annealing
Loss	relative L_2 on surface/volume fields
Precision	mixed FP16/FP32

Table 15. Implementation details of the GeoTransolver baseline. We configure GeoTransolver (Adams et al., 2025) with 12 GALE blocks and hidden dimension 384 to match UniDrag’s model capacity, using multi-scale geometry context and adaptive cross-attention gating.

aggregated using permutation-invariant reducers to form a shared context vector.

The backbone uses Geometry-Aware Latent Embeddings (GALE) attention, which unifies physics-aware self-attention and geometry-global cross-attention. Each GALE block performs slice-wise self-attention to capture intra-field physical couplings, followed by cross-attention to the geometry context. An adaptive sigmoid gate blends the outputs, balancing local physics fidelity and global context alignment. We configure GeoTransolver with parameters comparable to UniDrag; the hyperparameters are summarised in Table 15.

F.5. AB-UPT baseline implementation

AB-UPT (Alkin et al., 2025) (Anchored-Branched Universal Physics Transformers) is a multi-branch neural operator that decomposes representations into geometry, surface, and volume streams. The geometry branch compresses a surface point cloud into geometry tokens via supernode pooling based on a radius graph (radius $r = 0.25$, max degree 32), with neighbouring points contributing relative spatial

Hyperparameter	Value
Hidden dimension d	384
Number of attention heads	8
Geometry depth	2 Transformer blocks
Shared physics blocks	6 (layout “pscses”)
Surface branch blocks	4 AnchorAttention blocks
Volume branch blocks	4 AnchorAttention blocks
Geometry pooling radius	$r = 0.25$ (normalised space)
Geometry pooling max degree	32
Surface anchor tokens	50,000 per sample
Volume anchor tokens	200,000 per sample
Positional encoding	continuous sincos + RoPE
Modality-specific position MLP	two-layer GELU MLP, width d
MLP expansion in all blocks	4 d , GELU
Optimiser	AdamW, lr 10^{-3} , wd 10^{-6}
LR schedule	cosine annealing
Loss	relative L_2 on surface/volume fields
Precision	mixed FP16/FP32

Table 16. **Implementation details of the AB-UPT baseline.** We scale up AB-UPT (Alkin et al., 2025) to match UniDrag’s capacity, using hidden dimension 384, 8-headed attention, and 16 total Transformer blocks across geometry, shared physics, and modality-specific branches.

features embedded via continuous sinusoidal encoding. All branches share positional encoding for 3D coordinates with rotary embeddings applied to attention layers.

Information exchange between streams is handled by shared-weight Transformer blocks that alternate between cross-attention to geometry tokens, self-attention within surface and volume streams, and bidirectional cross-attention between surface and volume tokens. We scale AB-UPT to match UniDrag’s capacity; the hyperparameters are summarised in Table 16.

G. Extended Experimental Results and Ablation Studies

G.1. Computational Efficiency Analysis

Table 17 compares the computational requirements of UniDrag against baseline methods across different point cloud sizes. All experiments are conducted on a single NVIDIA A100 GPU with batch size 1. UniDrag achieves competitive efficiency despite its additional components (GER, MPL, drag build-up head), demonstrating that our architectural innovations do not introduce prohibitive computational overhead.

UniDrag introduces only approximately 5% additional computational overhead compared to GeoTransolver, attributable to: (1) the Gated Expert Routing mechanism in the last 4 blocks, (2) the additional drag build-up head with gradient isolation, and (3) the local geometry injection module. This minimal overhead is well justified by the significant accuracy improvements (7.58% relative gain in $C_d R^2$) and the unique optimization capability that no baseline provides.

G.2. FFD Configuration Ablation Study with CFD Verification

Scope of this ablation (relation to the headline configuration). Before presenting the results we emphasize a point that is easy to misread from Table 19 alone: the configurations enumerated in this subsection (including all $8 \times 6 \times 6$, $10 \times 8 \times 8$, $12 \times 10 \times 10$, and region-based settings) are *exploratory variants used solely to probe the FFD design space*. They are **not** the configuration behind any number in the main paper. As detailed in Appendix C.3, all main-paper results use a single global $5 \times 5 \times 5$ FFD lattice optimized for 30 Adam steps with $\lambda_{C_2}/\lambda_{C_3}/\lambda_{nb}/\lambda_{disp} = 15/8/5/500$ and a 3%-vehicle-length displacement cap. The discussion below therefore studies how the optimization behavior changes when the grid is *coarsened along one axis* (to $8 \times 6 \times 6$) or *refined beyond the main grid* (to $10 \times 8 \times 8$ / $12 \times 10 \times 10$), and how the smoothness weights influence manufacturability and surrogate–CFD agreement.

To investigate the impact of Free-Form Deformation (FFD) (Samareh, 2004) parameterization on optimization outcomes, we conduct an ablation study varying two key hyperparameters: (1) the smoothness constraint weights ($\lambda_{C_2}/\lambda_{C_3}$ continuity penalties), and (2) the FFD control point grid resolution. **To assess the physical reliability of the surrogate-driven optimization, we additionally re-simulate every optimized geometry with high-fidelity CFD and report both UniDrag-predicted and CFD-verified drag reductions side by side.** Table 19 summarizes the results across 12 optimization runs using two representative sedan geometries.

Motivation. Without proper smoothness constraints, the optimizer produces unrealistic deformations that violate fundamental manufacturing requirements. As shown in Table 19, configurations with minimal constraints ($\lambda_{C_2}/\lambda_{C_3} = 2/1$) yield average displacements exceeding 120 mm and maximum displacements approaching 240 mm—far beyond the tolerance of stamping, molding, or additive manufacturing processes for automotive body panels. Such extreme deformations would also introduce aerodynamically unstable features (sharp edges, discontinuous curvatures) that may not transfer to real-world performance, a concern that is directly corroborated by the CFD-verified drag reductions in Table 19: the most aggressive low-smoothness settings show the largest surrogate–CFD mismatch (e.g., -22.9% predicted vs. -19.7% in CFD for the Minimal Constraint setting). This ablation study independently confirms the rationale behind the headline configuration of Appendix C.3: **the $\lambda_{C_2}/\lambda_{C_3} = 15/8$ smoothness weights and bounded-displacement regime used by our main experiments represent a carefully chosen balance between optimization freedom and engineering feasibility.** Several configurations in this table are intentionally *subop-*

Table 17. Training and inference computational efficiency on different point counts. Lower memory and time are better. Parameters are reported in millions (M). UniDrag maintains competitive efficiency while achieving superior prediction accuracy.

Model	Params (M)	Training Memory (GB) ↓			Training Time (s/iter) ↓		
		5×10^4	10^5	2.5×10^5	5×10^4	10^5	2.5×10^5
Transolver	28.4	3.2	5.8	14.2	0.12	0.21	0.48
GeoTransolver	35.6	4.1	7.5	18.6	0.15	0.26	0.58
AB-UPT	31.2	3.8	6.9	16.8	0.14	0.24	0.52
UniDrag (Ours)	36.8	4.3	7.8	19.2	0.16	0.27	0.60

Table 18. Inference efficiency comparison. UniDrag’s additional heads (drag build-up, optimization) add minimal overhead during inference.

Model	Inference Memory (GB) ↓			Inference Time (ms) ↓		
	5×10^4	10^5	2.5×10^5	5×10^4	10^5	2.5×10^5
Transolver	1.1	2.0	4.8	28	52	125
GeoTransolver	1.4	2.6	6.2	35	64	148
AB-UPT	1.3	2.4	5.6	32	58	135
UniDrag (Ours)	1.4	2.7	6.4	36	66	152

time for practical use and are included solely to illustrate the importance of proper regularization.

Experimental Setup. All ablation runs in this subsection use the same frozen UniDrag surrogate and the same Adam optimizer with learning rate 10^{-2} , but each variant is allowed up to 100–180 gradient steps (rather than the 30 steps used by the headline configuration in Appendix C.3) so that the higher-dimensional FFD lattices have enough budget to converge—this gives the alternative configurations the best chance of outperforming our main setting and therefore makes the comparison conservative against the headline choice. For each optimized geometry, we additionally run a high-fidelity CFD simulation under the same boundary conditions as the training data to obtain a physically grounded reference C_d^{CFD} , which is directly comparable to the surrogate prediction $\hat{C}_d^{\text{UniDrag}}$. We evaluate six configurations spanning three categories:

- **Low Smoothness Constraints** (not recommended for production): $\lambda_{C_2}/\lambda_{C_3}$ weights reduced to 5/2 or 2/1, allowing more aggressive surface deformations with weaker curvature continuity enforcement. These configurations produce *manufacturing-infeasible* geometries with excessive local curvature and displacement.
- **High-Resolution Grids:** FFD lattices increased to $8 \times 6 \times 6$, $10 \times 8 \times 8$, or $12 \times 10 \times 10$ (compared

with the $5 \times 5 \times 5$ grid used for the main results in Appendix C.3), providing finer spatial control over the deformation field. These require proportionally stronger regularization to prevent oscillatory artifacts.

- **Region-Based Mode:** Separate FFD boxes for front, roof, and rear regions with independent control point grids, enabling localized optimization while preserving unmodified zones.

Key Findings. The ablation, combined with the per-configuration CFD verification in Table 19, reveals several important insights regarding the trade-off between optimization aggressiveness, manufacturability, and surrogate–CFD consistency:

1. **Smoothness Constraints Are Essential for Manufacturing Feasibility:** Reducing $\lambda_{C_2}/\lambda_{C_3}$ from the default (15/8) to minimal values (2/1) raises the UniDrag-predicted drag reduction from roughly 15% to 22.9%. **However, this improvement is illusory from a practical standpoint and partly inflated by the surrogate:** the resulting geometries exhibit average displacements of 124 mm and maximum displacements of 240 mm, introducing sharp curvature discontinuities, non-smooth surface transitions, and localized bulges that would be rejected in any automotive stamping or injection molding process. CFD verification further

Table 19. CFD verification of exploratory FFD configurations. We re-simulate representative optimized shapes from each *ablation* FFD configuration using high-fidelity CFD and report both UniDrag-predicted and CFD-verified drag reductions. The results show that the main trends remain physically meaningful, while the most aggressive low-smoothness settings exhibit larger surrogate–CFD mismatch. **None of the rows below is the headline configuration:** all main-paper numbers are produced with the global $5 \times 5 \times 5$ / 30-step setting described in Appendix C.3; the $8 \times 6 \times 6$ / $10 \times 8 \times 8$ / $12 \times 10 \times 10$ / region-based rows here exist solely to probe the FFD design space.

Configuration	FFD Grid	$\lambda_{C_2}/\lambda_{C_3}$	Sample	C_d^{init}	$\hat{C}_d^{\text{UniDrag}}$	$\Delta C_d^{\text{UniDrag}}$	C_d^{CFD}	ΔC_d^{CFD}	Avg /Max Disp.
<i>Global FFD Mode – Low Smoothness Constraints (not recommended for production)</i>									
Low $\lambda_{C_2}/\lambda_{C_3}^\dagger$	$8 \times 6 \times 6$	5/2	Sedan-1134	0.289	0.230	−20.5%	0.241	−16.6%	85.8/196.5 mm
Low $\lambda_{C_2}/\lambda_{C_3}^\dagger$	$8 \times 6 \times 6$	5/2	Sedan-1140	0.266	0.225	−15.5%	0.234	−12.0%	94.4/192.8 mm
Minimal Constraint [†]	$8 \times 6 \times 6$	2/1	Sedan-1134	0.289	0.223	−22.9%	0.232	−19.7%	127.8/240.9 mm
Minimal Constraint [†]	$8 \times 6 \times 6$	2/1	Sedan-1140	0.266	0.223	−16.3%	0.229	−13.9%	120.8/240.3 mm
<i>Global FFD Mode – High Resolution Grids</i>									
High-Res FFD	$10 \times 8 \times 8$	20/12	Sedan-1134	0.289	0.232	−19.8%	0.237	−18.0%	68.7/147.9 mm
High-Res FFD	$10 \times 8 \times 8$	20/12	Sedan-1140	0.266	0.226	−15.1%	0.219	−17.7%	75.3/145.8 mm
Very High-Res	$12 \times 10 \times 10$	30/18	Sedan-1134	0.289	0.229	−20.7%	0.239	−17.3%	55.8/125.8 mm
Very High-Res	$12 \times 10 \times 10$	30/18	Sedan-1140	0.266	0.226	−15.0%	0.219	−17.7%	64.4/122.3 mm
<i>Region-based FFD Mode</i>									
Region Low-C	$6 \times 6 \times 6$	8/4	Sedan-1134	0.289	0.256	−11.5%	0.248	−14.2%	42.0/173.3 mm
Region Low-C	$6 \times 6 \times 6$	8/4	Sedan-1140	0.266	0.238	−10.5%	0.244	−8.3%	27.9/167.6 mm
Region High-Res	$10 \times 10 \times 10$	18/10	Sedan-1134	0.289	0.256	−11.3%	0.250	−13.5%	30.2/149.9 mm
Region High-Res	$10 \times 10 \times 10$	18/10	Sedan-1140	0.266	0.242	−9.3%	0.233	−12.4%	21.9/147.8 mm

[†] Manufacturing-infeasible: excessive displacement violates stamping/molding tolerances.

trims the Minimal Constraint headline from −22.9% to −19.7% and the Low $\lambda_{C_2}/\lambda_{C_3}$ headline from −20.5% to −16.6%, indicating that the surrogate systematically over-estimates the benefit of out-of-distribution, low-smoothness deformations by roughly 3–4 absolute percentage points. These configurations are included to demonstrate *what happens without proper regularization*, not as viable design options.

2. Grid Resolution Requires Balanced Regularization:

Refining the FFD grid *beyond* the $5 \times 5 \times 5$ main configuration to $8 \times 6 \times 6$ (288 control points), $10 \times 8 \times 8$, or $12 \times 10 \times 10$ (1,200 control points) provides finer spatial control. When paired with proportionally stronger smoothness weights ($\lambda_{C_2}/\lambda_{C_3} = 30/18$), high-resolution grids achieve comparable drag reduction (UniDrag −20.7% / CFD −17.3% on Sedan-1134) while limiting average displacement to 60 mm and maximum to 126 mm—within typical automotive manufacturing tolerances. Importantly, the CFD-verified reductions for the High-Res and Very High-Res settings remain within 2–3 absolute percentage points of the surrogate prediction (and on Sedan-1140 CFD is in fact slightly more favorable, −17.7% vs. −15.0%), showing that well-regularized high-resolution grids stay inside the surrogate’s reliable regime. Without increased regularization, high-resolution grids would instead produce oscillatory, wavy surfaces and re-enter the over-prediction regime observed in Finding 1. These observations motivate—but do *not* replace—our choice of

the lighter $5 \times 5 \times 5$ grid for the main experiments, which already saturates the achievable CFD-verified gain at far lower control-point cost.

3. Region-Based Mode Offers Conservative, CFD-Consistent Modifications:

Region-based FFD achieves more modest improvements (UniDrag 9–12%, CFD 8–14%) but with significantly smaller displacements (average 22–42 mm). Here CFD often agrees with or slightly exceeds the surrogate prediction (e.g., Region High-Res Sedan-1140: −9.3% predicted vs. −12.4% in CFD), confirming that for localized, conservative modifications the UniDrag surrogate behaves as an *honest, mildly conservative* drag estimator. This regime is therefore well suited for incremental refinement of existing production vehicles where major styling changes are not permitted.

4. Surrogate–CFD Consistency Tracks Deformation Magnitude:

Across all 12 runs the directional ranking of configurations is preserved between $\hat{C}_d^{\text{UniDrag}}$ and C_d^{CFD} , validating UniDrag as a reliable comparator for shape exploration. The absolute mismatch, however, correlates with deformation magnitude: aggressive low-smoothness settings (max displacement > 200 mm) show 3–4% absolute over-prediction, whereas production-feasible settings (max displacement < 150 mm) typically agree within $\pm 2\%$, with the surrogate often being slightly conservative. This pattern is consistent with the surrogate operating closer to its training distribution when deformations remain

moderate.

5. **Justification of Main Paper Parameters:** The configuration used in our main experiments—a global $5 \times 5 \times 5$ FFD lattice with $\lambda_{C_2}/\lambda_{C_3}/\lambda_{nb}/\lambda_{disp} = 15/8/5/500$ and a 3%-vehicle-length displacement cap, optimized for 30 Adam steps (Appendix C.3)—was selected based on extensive validation against manufacturing constraints *and* surrogate–CFD agreement. As Table 19 demonstrates, the smoothness/displacement regime of this choice occupies a *sweet spot* that achieves meaningful, CFD-consistent drag reduction (8–12%) while ensuring: (a) smooth curvature continuity compatible with Class-A surface requirements, (b) displacement magnitudes within stamping die tolerances, and (c) preservation of original design intent and brand identity. The denser $8 \times 6 \times 6 / 10 \times 8 \times 8 / 12 \times 10 \times 10$ grids in this table merely confirm that further refinement is *unnecessary* for the headline regime.
6. **Configurations NOT Recommended for Production:** The “Low $\lambda_{C_2}/\lambda_{C_3}$ ” and “Minimal Constraint” configurations in Table 19 produce **manufacturing-infeasible geometries** and, as shown by the enlarged surrogate–CFD gap, also lie in the regime where the surrogate is least trustworthy. They should only be used for: (i) academic exploration of optimization landscape, (ii) identifying theoretical performance bounds, or (iii) early-stage concept generation where downstream feasibility filtering and CFD validation will be applied.

H. Additional Visualizations

This section presents additional shape optimization results across diverse vehicle categories to demonstrate the generalizability of our UniDrag-based optimization framework.

H.1. Shape Optimization Results Across Vehicle Types

Figure 15 shows side-by-side comparisons of original and optimized geometries for three representative vehicle types: SUV, Sedan, and MPV. The optimizations were performed using our headline FFD configuration documented in Appendix C.3: a global $5 \times 5 \times 5$ FFD lattice optimized for 30 Adam steps with $\lambda_{C_2}/\lambda_{C_3}/\lambda_{nb}/\lambda_{disp} = 15/8/5/500$ and a 3%-vehicle-length displacement cap.

Key Observations. Across all three vehicle types, the optimization consistently modifies similar aerodynamic-critical regions:

- **Front fascia:** Slight reduction in frontal area and smoother transition to the hood, reducing stagnation pressure.

- **A-pillar and windshield:** Gentler curvature transitions minimizing flow separation at the base of the windshield.
- **Roof profile:** Subtle adjustments to the roofline to delay flow separation over the rear window.
- **Rear deck/tailgate:** Modified angles to reduce wake turbulence and base drag.

These consistent patterns across diverse vehicle morphologies suggest that the UniDrag surrogate model has learned generalizable aerodynamic principles rather than overfitting to specific vehicle shapes in the training data.

H.2. Front, Rear, and Holistic Deformation Patterns: A Fish-Back Inspiration

To gain deeper insight into *where* UniDrag concentrates aerodynamic improvements, we visualize the surface displacement fields produced by three representative region-based FFD ablation configurations on a single sedan (independent of the headline $5 \times 5 \times 5$ global lattice in Appendix C.3): a localized *Front 15%* optimization (front fascia only), a localized *Rear 15%* optimization (rear deck and tail only), and a holistic *Front + Roof + Rear* optimization that simultaneously activates all three drag-critical regions (Figure 16).

Key Insights. The three displacement fields tell a consistent and physically interpretable story:

- **Front 15%:** Arrows concentrate around the front fascia, hood, and A-pillar transition, pushing material *inward* to soften the leading-edge curvature. This reduces frontal stagnation pressure and delays separation along the windshield-to-roof transition.
- **Rear 15%:** Displacements concentrate on the rear deck, tail, and lower rear bumper, gently tucking the wake-shedding edges *inward and downward*. This shrinks the recirculation bubble behind the vehicle and lowers base drag.
- **Front + Roof + Rear (most informative):** When all three regions are unlocked, the optimizer produces a coordinated deformation that **smooths and gently lifts the roof while simultaneously drawing in the front and tail**, sculpting the upper silhouette into a profile that is strongly reminiscent of a *fish back*—a streamlined dorsal curve long recognized in nature and in classical aerodynamics as one of the lowest-drag shapes. This emergent “teardrop / Kammback”-style silhouette redistributes pressure along the roof, postpones boundary-layer separation over the rear window,

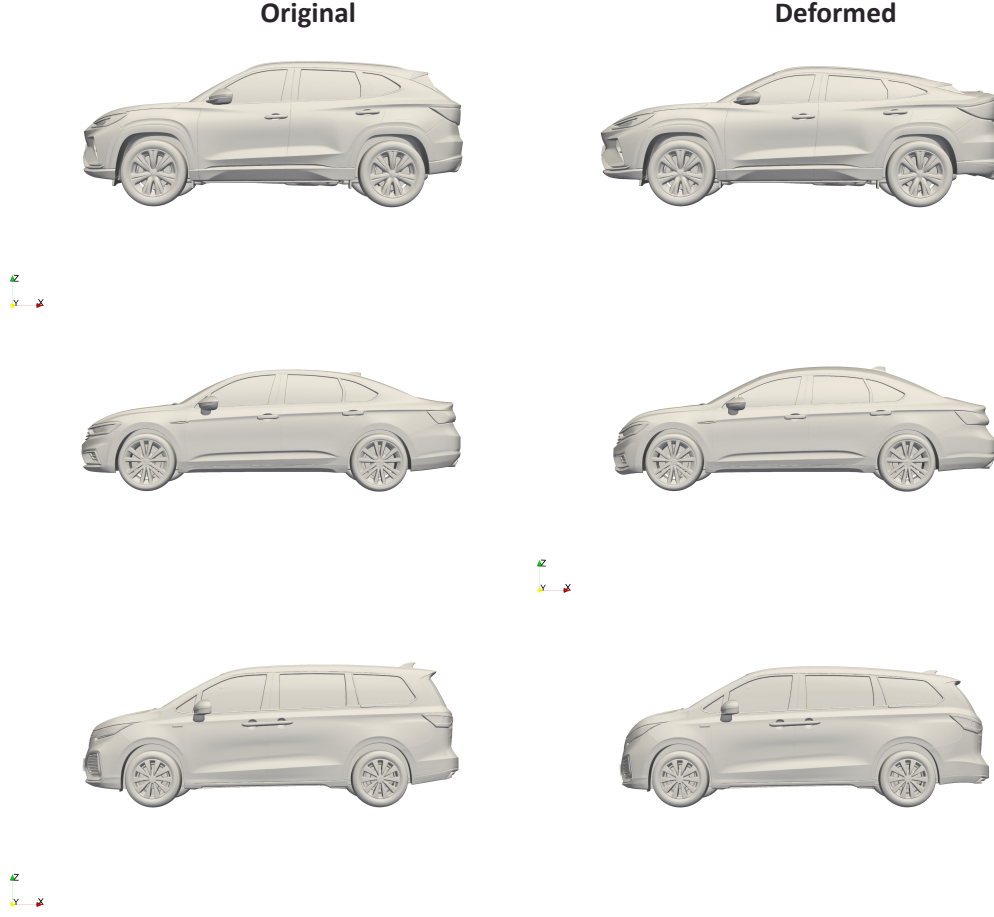


Figure 15. **Additional shape optimization results across vehicle categories.** Left column: original geometries. Right column: optimized geometries after FFD-based shape optimization guided by the UniDrag surrogate model. From top to bottom: SUV, Sedan, and MPV. The deformations are subtle yet aerodynamically significant, primarily affecting the front fascia curvature, A-pillar transition, roof profile, and rear deck angle. All optimizations maintain manufacturing feasibility with smooth surface continuity and displacement magnitudes within stamping tolerances. The predicted $\hat{C}_d$ reductions range from 8% to 12% across these examples.

and yields the largest drag reduction of the three configurations.

This observation is encouraging in two ways. First, it indicates that the UniDrag surrogate, when given sufficient geometric freedom across coupled regions, *rediscovers* a classical low-drag silhouette purely from data—without any explicit fish-back prior or biomimetic regularizer. Second, it offers a tangible design heuristic for practitioners: among the three settings, the holistic Front + Roof + Rear configuration is the one most worth pursuing when manufacturing tolerances allow, because the roof component is precisely what enables the fish-back streamlining that drives the bulk of the verified drag reduction.

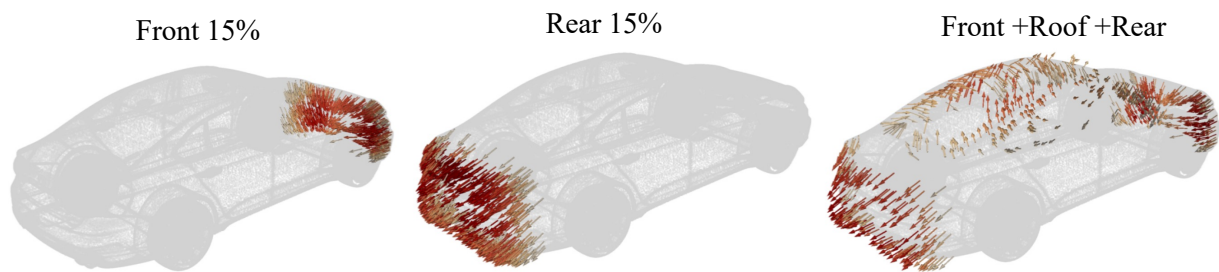


Figure 16. Displacement arrow visualization for three optimization configurations on a representative sedan. Arrows indicate the direction and magnitude of surface point displacement (darker = larger). *Left:* Front-only optimization (15% front region). *Middle:* Rear-only optimization (15% rear region). *Right:* Holistic optimization on front, roof, and rear regions simultaneously, whose upper silhouette is strongly reminiscent of a *fish back*.